PHYSICS SSC–II

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Time allowed: 2:40 Hours **Total Marks Sections B and C: 53**

SECTION – B (Marks 33)

Q. 2 Answer the following parts briefly. (11 x 3 = 33)

سوال نمبر2 درج ذیل اجزاء کے مختصر جوابات تحریر کریں۔

(i)	State the melting point and boiling point of water at standard atmospheric pressure. معیاری فضائی دباؤ پر پانی کا نقطہ پگھلاؤ اور نقطہ ابال بیان کریں۔	1.5 + 1.5	OR	State the main regions of the electromagnetic spectrum in order of increasing frequency. برقی مقناطیسی طیف کے اہم خطوں کو بڑھتے ہوئے تعدد کی ترتیب سے بیان کریں۔	1+2
(ii)	Briefly describe latent heat and write its SI unit. حرارت مخفی (Latent Heat) کی مختصر وضاحت کریں اور اس کی SI اکائی لکھیں۔	2+1	OR	Briefly describe the half-life of a radioactive substance and write its formula. کسی تابکار مادے کی نصف عمر کی مختصر وضاحت کریں اور اس کا فارمولا لکھیں۔	2+1
(iii)	How do land and sea breezes form? خشکی اور سمندری ہوائیں کیسے بنتی ہیں؟	1.5 + 1.5	OR	How is knowledge of the properties of sound waves applied in the design of buildings with respect to acoustics? صوتی لہروں کی خصوصیات کے علم کو عمارتوں کے ڈیزائن میں صوتیات کے حوالے سے کیسے استعمال کیا جاتا ہے؟	1+2
(iv)	Explain melting in terms of energy transfer without a change in temperature. درجہ حرارت میں تبدیلی کے بغیر توانائی کی منتقلی کے لحاظ سے پگھلنے کے عمل کی وضاحت کریں۔	1+2	OR	Explain electrical breakdown in a gas. گیس میں برقی شکست کی وضاحت کریں۔	1+2
(v)	Explain what is meant by the apparent expansion of a liquid. مائع کے ظاہری پھیلاؤ سے کیا مراد ہے؟ وضاحت کریں۔	1+2	OR	State the rule that relates the currents entering and leaving a junction in a parallel circuit. وہ اصول بیان کریں جو متوازی سرکٹ میں کسی جنکشن میں داخل ہونے اور اس سے نکلنے والی برقی رووں کے درمیان تعلق کو ظاہر کرتا ہے۔	1+2
(vi)	Describe the position and two characteristics of the image formed by a plane mirror. مستوی آئینے سے بننے والے عکس کی پوزیشن اور دو خصوصیات بیان کریں۔	1+2	OR	How can a conductor be charged by electrostatic induction and then earthed? Explain briefly. کسی موصل کو برقی سکونی القا کے ذریعے کیسے چارج کیا جا سکتا ہے اور پھر اسے ارتھ کیسے کیا جاتا ہے؟ مختصر وضاحت کریں۔	1+2
(vii)	How do changes in amplitude and frequency affect the loudness and pitch of a sound wave? ایمپلی ٹیوڈ اور تعدد میں تبدیلیاں صوتی لہر کی بلندی آواز (Loudness) اور زیر و بم (Pitch) کو کیسے متاثر کرتی ہیں؟	1.5 + 1.5	OR	Explain the dispersion of light by a prism. منشور (Prism) کے ذریعے روشنی کے انتشار کی وضاحت کریں۔	1+2
(viii)	Describe how waves undergo reflection, refraction and diffraction. بیان کریں کہ لہروں میں انعکاس، انعطاف اور انحراف کیسے واقع ہوتے ہیں۔	1+1 +1	OR	Describe the pattern of the magnetic field around a straight current-carrying wire. سیدھے برقی رو بردار تار کے گرد مقناطیسی میدان کے نمونے کو بیان کریں۔	1+2
(ix)	Light travels from glass (n = 1.5) towards air. Calculate the critical angle for this glass–air boundary. روشنی شیشے (n = 1.5) سے ہوا کی طرف سفر کرتی ہے۔ شیشہ-ہوا سطح کے لیے کریٹکل اینگل معلوم کریں۔	1+2	OR	A current of 2 A flows through a 6 Ω resistor. Calculate the potential difference (p.d.) across it. ایک 6 Ω مزاحمت کار میں 2 A برقی رو بہتی ہے۔ اس کے سروں کے درمیان فرق پوٹینشل (p.d.) معلوم کریں۔	1+2
(x)	Briefly describe the resistivity of a conducting wire and give its SI unit. کسی موصل تار کی مخصوص مزاحمت کی مختصر وضاحت کریں اور اس کی SI اکائی بتائیں۔	1+2	OR	Why must the outer casing of an electrical appliance be either non-conducting (double-insulated) or earthed? کسی برقی آلے کا بیرونی غلاف یا تو غیر موصل یا ارتھ شدہ ہونا کیوں ضروری ہے؟	1+2
(xi)	A transformer has 100 turns on its primary coil and 400 turns on its secondary coil. If the primary voltage is 20 V, find the secondary voltage. ایک ٹرانسفارمر کی پرائمری کوائل میں 100 چکر اور سیکنڈری کوائل میں 400 چکر ہیں۔ اگر پرائمری وولٹیج 20 V ہو تو سیکنڈری وولٹیج معلوم کریں۔	1+2	OR	Write the nuclide decay equation for the alpha emission of Uranium U_{92}^{239} (which decays into Thorium, Th). یورینیم U_{92}^{239} کے الفا اخراج کے نتیجے میں تھوریم (Th) بننے کی نیوکلائیڈ تحلیل مساوات لکھیں۔	1+2

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SECTION – C (Marks 20)

Note: Answer the following questions. (4 x 5=20) درج سوالات کے جوابات تحریر کریں۔

Q.3	Describe the action of a thin converging lens on a parallel beam of light and draw a labelled ray diagram to show the formation of a real image of an object placed beyond its focal length. ایک باریک کنورجنگ لینز کے متوازی شعاعی بیم پر عمل کو بیان کریں اور فوکل لمبائی سے باہر رکھے گئے جسم کا حقیقی عکس بننے کا عمل دکھانے کے لیے نامزد شعاعی خاکہ (Labelled Ray Diagram) بنائیں۔	3+2	OR	Describe the operation of a simple d.c. electric motor, including the role of the split-ring, commutator and brushes. ایک سادہ ڈی سی برقی موٹر کے کام کرنے کا طریقہ بیان کریں، جس میں اسپلٹ رنگ، کمیوٹیٹر اور برشز کے کردار کو بھی شامل کریں۔	2+1 +1 +1
Q.4	Calculate the heat energy required to raise the temperature of 2 kg of water from 20°C to 80°C. (Specific heat capacity of water = 4200 J/kg°C) 2 kg پانی کا درجہ حرارت 20°C سے 80 °C تک بڑھانے کے لیے درکار حرارتی توانائی معلوم کریں۔ (Specific heat capacity of water = 4200 J/kg°C)	05	OR	Calculate the cost of running a 2000 W electric heater for 5 hours if the cost of electricity is Rs. 15 per kWh. اگر بجلی کی قیمت 15 روپے فی kWh ہو تو 2000 W کے برقی ہیٹر کو 5 گھنٹے چلانے کی لاگت معلوم کریں۔	05
Q.5	Explain critical angle and total internal reflection. State the necessary conditions for total internal reflection to occur. زاویۂ حد اور مکمل داخلی انعکاس کی وضاحت کریں۔ مکمل داخلی انعکاس واقع ہونے کے لیے ضروری شرائط بیان کریں۔	1+1 +3	OR	Differentiate between analogue and digital electronics. What makes a digital system more reliable than an analogue system? اینالاگ اور ڈیجیٹل الیکٹرانکس میں فرق بیان کریں۔ ڈیجیٹل نظام کو اینالاگ نظام کی نسبت زیادہ قابلِ اعتماد بنانے والی خصوصیت کیا ہے؟	1+1 +3
Q.6	Describe the working of a diode, including how it behaves under forward and reverse biasing. ڈائیوڈ کے کام کرنے کا طریقہ بیان کریں، نیز وضاحت کریں کہ یہ فارورڈ بائسنگ اور ریورس بائسنگ میں کس طرح عمل کرتا ہے؟	1+2 +2	OR	Describe nuclear fusion with a suitable example, and explain why it is the source of energy in stars such as the Sun. مناسب مثال کے ساتھ نیوکلیائی فیوژن کو بیان کریں اور وضاحت کریں کہ یہ سورج جیسے ستاروں میں توانائی کا ذریعہ کیوں ہے؟	1+1 +3

- $V = IR$
- $R = \rho \frac{L}{A}$
- $Q = mc\Delta T$
- $n = \frac{1}{\sin c}$
- $\frac{V_s}{V_p} = \frac{N_s}{N_p}$
- $\text{Cost} = \left(\frac{\text{Power} \times \text{time}}{1000}\right) \times (\text{rate per hour})$

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PHYSICS
SSC-II / CLASS 10

FBISE • 2026
Four-paper question-and-answer compilation

Prepared for Fakhar STEM Sphere

One 2nd Annual paper + three 1st Annual sets, in upload order.

Inside: every available question followed by its solution; all visible OR alternatives; worked numericals; reasoning for MCQs; labelled diagrams; logic-gate truth tables.

Coverage of the supplied images

Paper	Material supplied and solved	Source gap
1. 2nd Annual	Section B: (i)–(xi), all ORs; Section C: Q.3–Q.6, all ORs.	Section A / MCQs were not uploaded.
2. 1st Annual - Set 1	MCQs 1–12; Section B: (i)–(xi), all ORs; Section C: Q.3–Q.6, all ORs.	None within the supplied set.
3. 1st Annual - Set 2	MCQs 1–12; Section B: (i)–(viii), all ORs; Section C: Q.3–Q.6, all ORs.	Section B (ix)–(xi) and their ORs were not uploaded.
4. 1st Annual - Set 3	MCQs 1–12; Section B: (i)–(xi), all ORs; Section C: Q.3–Q.6, all ORs.	None within the supplied set.

Set numbers distinguish the three 1st Annual papers in the order shared; morning/evening or regional labels were not visible. Session names follow the user's identification. Questions are faithfully transcribed into clear English; obvious printing/grammar errors are normalised without changing the task.

Important: This is an independently prepared study solution, not an official FBISE marking scheme. Answers cover the expected concepts, working and units; marks depend on the official scheme and the student's response. Unseen questions have not been invented.

How to use these answers

For a 3-mark answer, include a clear definition/law plus the requested explanation or example. For numericals, write given data, convert units, state the formula, substitute and box the answer with its unit. Label diagrams and show ray/current directions. In an exam, attempt the required side of each OR; both are included here for practice.

Scientific review: The weak MCQ about sound and density is flagged; the convex-lens numerical states its real-image assumption; latent heat is distinguished from specific latent heat; magnitude/sign conventions are explained where the question is ambiguous.

Paper 1 | 2nd Annual Examination 2026

Source: the two full-page photographs, 1(1).jpeg and 2.jpeg. All Section B and C questions are solved. The MCQ page is absent from the uploads.

Section B | Q.2: short answers (11 × 3 = 33)

Q.2 (i) State the melting point and boiling point of water at standard atmospheric pressure.

At standard atmospheric pressure (approximately 1.013×10^5 Pa), ice melts at **0 °C (273 K)** and water boils at **100 °C (373 K)**.

OR State the main regions of the electromagnetic spectrum in order of increasing frequency.

The electromagnetic spectrum is the complete range of electromagnetic waves. In **increasing frequency**:
Radio → Microwaves → Infrared → Visible → Ultraviolet → X-rays → Gamma rays.
Wavelength decreases in this order.

Q.2 (ii) Briefly describe latent heat and write its SI unit.

Latent heat is energy absorbed or released during a change of state **without a temperature change**. For example, ice at 0 °C absorbs energy to become water at 0 °C.
Q = mL. Latent heat Q is measured in **joules (J)**. Specific latent heat L is energy per kilogram and has unit **J kg^{-1}**.

OR Briefly describe the half-life of a radioactive substance and write its formula.

Half-life is the time required for half the undecayed radioactive nuclei in a sample to decay; its activity also falls to half.
$T_{½} = 0.693/\lambda$, where λ is the decay constant. The remaining quantity is **$N = N_0(½)^{t/T_{½}}$**.
Example: with an 8-day half-life, 40 g of radioactive iodine becomes 20 g after 8 days and 10 g after 16 days.

Q.2 (iii) How do land and sea breezes form?

Day: sea breeze. Land heats faster than seawater. Air above land warms, expands and rises. Cooler air flows **from sea to land** to replace it.
Night: land breeze. Land cools faster than the sea. Air above the relatively warmer sea rises, and cooler air flows **from land to sea**. Both are convection currents.

OR How is knowledge of sound-wave properties applied to building acoustics?

Acoustics is the study of sound and its behaviour. In halls, curtains, carpets, upholstered seats and acoustic panels absorb sound to control excessive reverberation and echoes. Suitable ceiling shapes and reflectors distribute sound evenly towards the audience. Large concave surfaces that focus sound at particular points are avoided, so speech remains clear.

Q.2 (iv) Explain melting in terms of energy transfer without a change in temperature.

Melting is the change from solid to liquid. At the melting point, supplied energy overcomes forces holding particles in fixed positions, increasing their potential energy. Their average kinetic energy does not increase, so temperature stays constant until melting is complete.
$\mathbf{Q = mL_f}$, where L_f is specific latent heat of fusion.

OR Explain electrical breakdown in a gas.

Electrical breakdown occurs when a normally insulating gas becomes conducting in a sufficiently strong electric field. Free electrons accelerate and collide with gas atoms, ionising them. These collisions create more electrons and positive ions, producing an avalanche of charge carriers and a large current. Sparks and lightning are examples.

Q.2 (v) Explain apparent expansion of a liquid.

Apparent expansion is the observed increase in a liquid's volume relative to its heated container. Both expand, so some actual liquid expansion occupies the extra container capacity.
$\mathbf{\Delta V_{apparent} = \Delta V_{real} - \Delta V_{container}}$.
For the same initial volume and temperature rise: $\mathbf{\beta_{apparent} = \beta_{liquid} - \beta_{container}}$.

OR State the rule relating currents entering and leaving a junction in a parallel circuit.

Kirchhoff's current law: the total current entering a junction equals the total current leaving it.
$\mathbf{\Sigma I_{in} = \Sigma I_{out}}$. For two branches, $\mathbf{I = I_1 + I_2}$.
This follows from conservation of electric charge: charge does not continuously accumulate at the junction.

Q.2 (vi) Describe the position and two characteristics of the image formed by a plane mirror.

The image forms **behind the plane mirror at the same perpendicular distance** as the object is in front. It is **virtual** (cannot be caught on a screen) and **upright and equal in size** to the object. It is also laterally inverted.

OR How can a conductor be charged by electrostatic induction and earthing?

Place a neutral metal conductor on an insulating stand. Bring a negatively charged rod close without touching. Electrons are repelled to the far side. Earth the conductor while retaining the rod; electrons flow to Earth. **Disconnect Earth first**, then remove the rod. The conductor remains **positively charged**, opposite to the inducing rod; its charge redistributes over its surface.

Q.2 (vii) How do changes in amplitude and frequency affect loudness and pitch?

Increasing sound-wave amplitude increases intensity and generally loudness: $\mathbf{I \propto A^2}$. Smaller amplitude gives softer sound. Increasing frequency raises pitch; decreasing frequency lowers pitch. Thus **amplitude mainly controls loudness, while frequency mainly controls pitch**. Hitting a drum harder makes it louder; a whistle has a higher pitch than a bass drum.

OR Explain dispersion of light by a prism.

Dispersion is the splitting of white light into its component colours by a prism. Glass has different refractive indices for different wavelengths, so colours refract through different angles. **Red deviates least and violet most.** The spectrum is red, orange, yellow, green, blue, indigo and violet.

Q.2 (viii) Describe reflection, refraction and diffraction of waves.

Reflection: waves bounce back into their original medium at a boundary; angle of incidence equals angle of reflection.
Refraction: wave speed and wavelength change on entering another medium; direction generally changes at oblique incidence, while frequency remains constant.
Diffraction: waves spread around obstacles or through openings, especially when their size is comparable to wavelength.

OR Describe the magnetic field around a straight current-carrying wire.

Field lines are **concentric circles** centred on the wire in planes perpendicular to it. By the **right-hand grip rule**, point the thumb along conventional current; curled fingers show field direction. The field is stronger for greater current and closer to the wire: **$B \propto I/r$**.

Q.2 (ix) Light travels from glass (n = 1.5) towards air. Calculate the critical angle.

Given: $n_1 = 1.5$; $n_2 \approx 1$
Formula: $\sin C = n_2/n_1$
Working: $\sin C = 1/1.5 = 0.6667$; $C = \sin^{-1}(0.6667)$
Answer: C ≈ 41.8° (measured from the normal inside glass).

OR A current of 2 A flows through a 6 Ω resistor. Calculate its potential difference.

Given: $I = 2$ A; $R = 6\ \Omega$
Formula: $V = IR$
Working: $V = 2 \times 6$
Answer: V = 12 V.

Q.2 (x) Describe the resistivity of a conducting wire and give its SI unit.

Resistivity measures how strongly a material opposes electric current. For a uniform wire, **$R = \rho L/A$**, hence **$\rho = RA/L$**. At fixed temperature it depends on the material, rather than the wire's length or area. Its SI unit is **ohm metre (Ω m)**.

OR Why must an appliance casing be double-insulated or earthed?

Double insulation: extra insulation, usually including a non-conducting outer casing, separates accessible parts from live conductors.
Earthing: a metal casing connects to Earth through a low-resistance wire. If a live wire touches it, fault current operates the fuse or circuit breaker, disconnecting the supply and reducing electric-shock risk.

Q.2 (xi) A transformer has 100 primary turns and 400 secondary turns. If the primary voltage is 20 V, find the secondary voltage.

Given: $N_p = 100$; $N_s = 400$; $V_p = 20$ V
Formula: $V_s/V_p = N_s/N_p$
Working: $V_s = 20 \times 400/100$
Answer: $V_s = 80$ V; it is a step-up transformer.

OR Write the alpha-decay equation for uranium-238 decaying into thorium.

In alpha decay, mass number falls by 4 and atomic number by 2:
$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$ + energy.
Mass numbers balance: 238 = 234 + 4. Atomic numbers balance: 92 = 90 + 2.

Section C | Long answers (4 × 5 = 20)

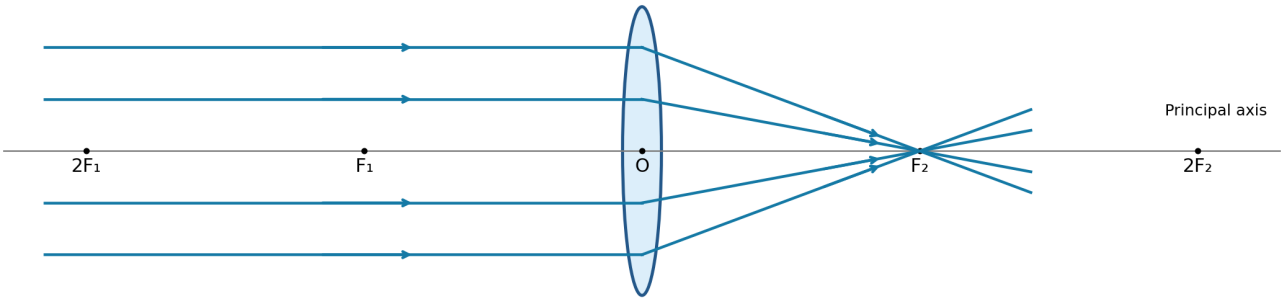
Q.3 Describe the action of a thin converging lens on a parallel beam. Draw a labelled ray diagram for a real image of an object beyond its focal length.

Parallel beam: A thin converging (convex) lens is thicker at the centre. Refraction at its surfaces makes rays parallel to the principal axis converge at the principal focus F. The distance OF from optical centre O is the focal length f.

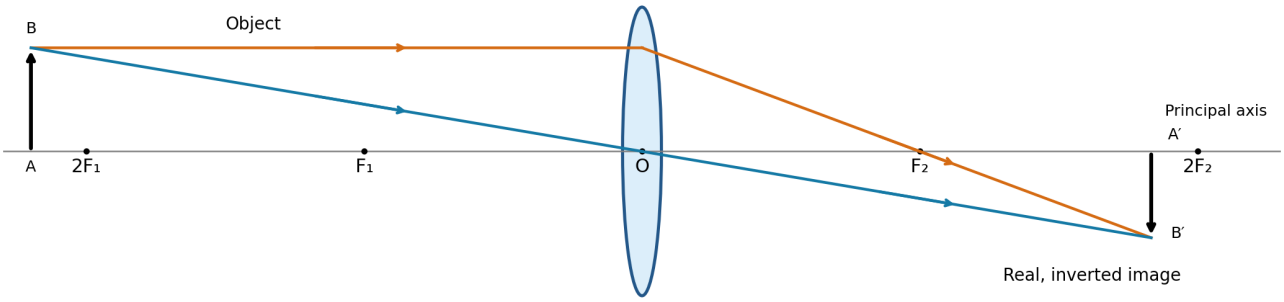
Real-image formation: Place the object at a distance greater than f. From its top draw (1) a ray parallel to the axis, which refracts through the far focus F_2, and (2) a ray through O, which continues without appreciable deviation. Their actual intersection locates the real, inverted image on the opposite side. It can be received on a screen.

Position and size: Beyond $2F_1$ gives an image between F_2 and $2F_2$, diminished; at $2F_1$ gives an equal-sized image at $2F_2$; between F_1 and $2F_1$ gives an enlarged image beyond $2F_2$. All three images are real and inverted.

A. Parallel rays converge at the principal focus



B. Real image: object beyond 2F₁ (one valid case of object beyond F₁)



Parallel rays converge at F_2. The example object is beyond $2F_1$; the image is between F_2 and $2F_2$, real, inverted and diminished.

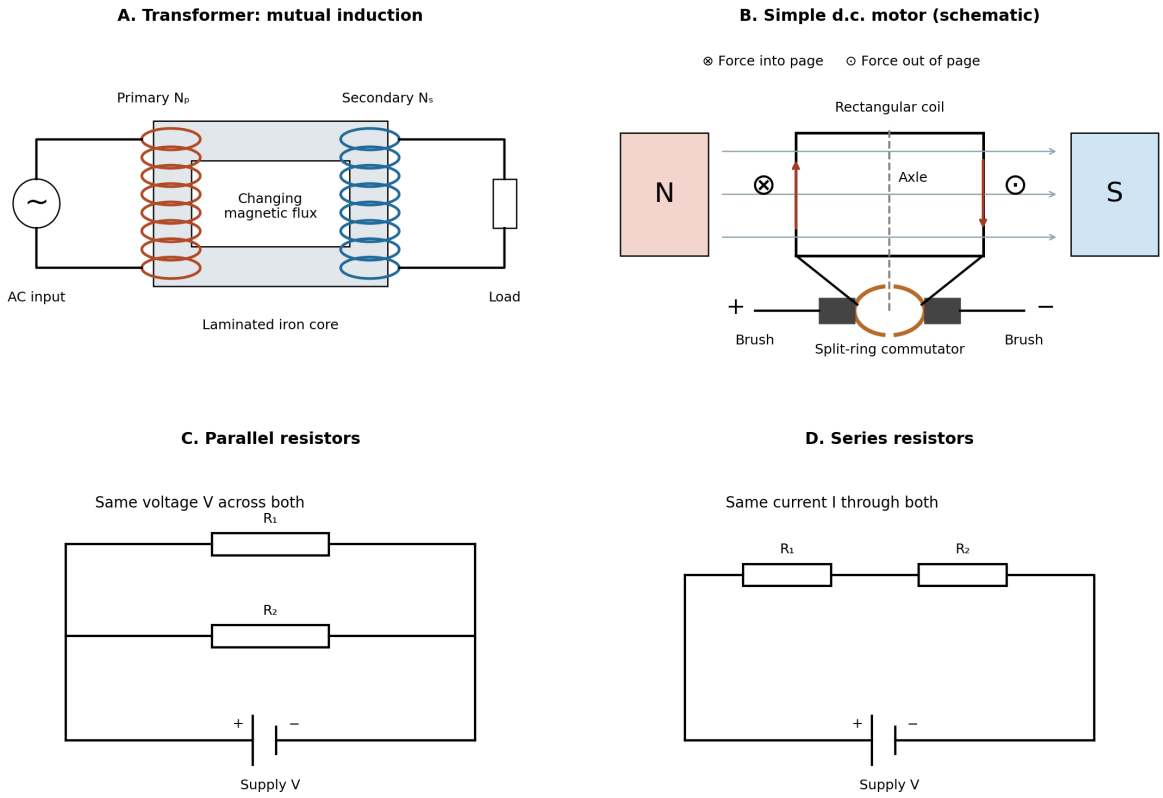
Q.3 OR Describe a simple d.c. motor, including the split-ring commutator and brushes.

Principle: A motor converts electrical energy into mechanical energy. A current-carrying conductor in a magnetic field experiences force; Fleming's left-hand rule gives its direction.

Construction: A rectangular insulated-wire coil rotates on an axle between magnetic poles. The coil ends connect to two insulated halves of a split-ring commutator. Stationary carbon brushes connect the commutator to a d.c. supply.

Working: Current flows in opposite directions in the two opposite sides of the coil. The field exerts opposite forces on them, forming a couple that turns the coil. After each half-turn, the split-ring commutator reverses the coil current. The forces reverse at the appropriate instant, keeping the turning effect in the same rotational sense. Momentum carries the coil through positions of zero turning effect.

Commutator: reverses coil current every half-turn to maintain rotation in one direction.
Brushes: maintain sliding electrical contact between stationary supply and rotating commutator. They do not themselves reverse the current.



Use panel B for the motor. Panel A is a transformer; C and D are resistor connections used later.

Q.4 Calculate heat needed to raise 2 kg of water from 20 °C to 80 °C, with c = 4200 J kg⁻¹ K⁻¹.

Given: m = 2 kg; c = 4200 J kg^{-1} K^{-1}; T_1 = 20 °C; T_2 = 80 °C
Formula: Q = mcΔT
Working: ΔT = 80 − 20 = 60 K; Q = 2 × 4200 × 60
Answer: Q = 504,000 J = 504 kJ (neglect heat losses).

Q.4 OR Calculate the cost of running a 2000 W heater for 5 hours at Rs. 15 per kWh.

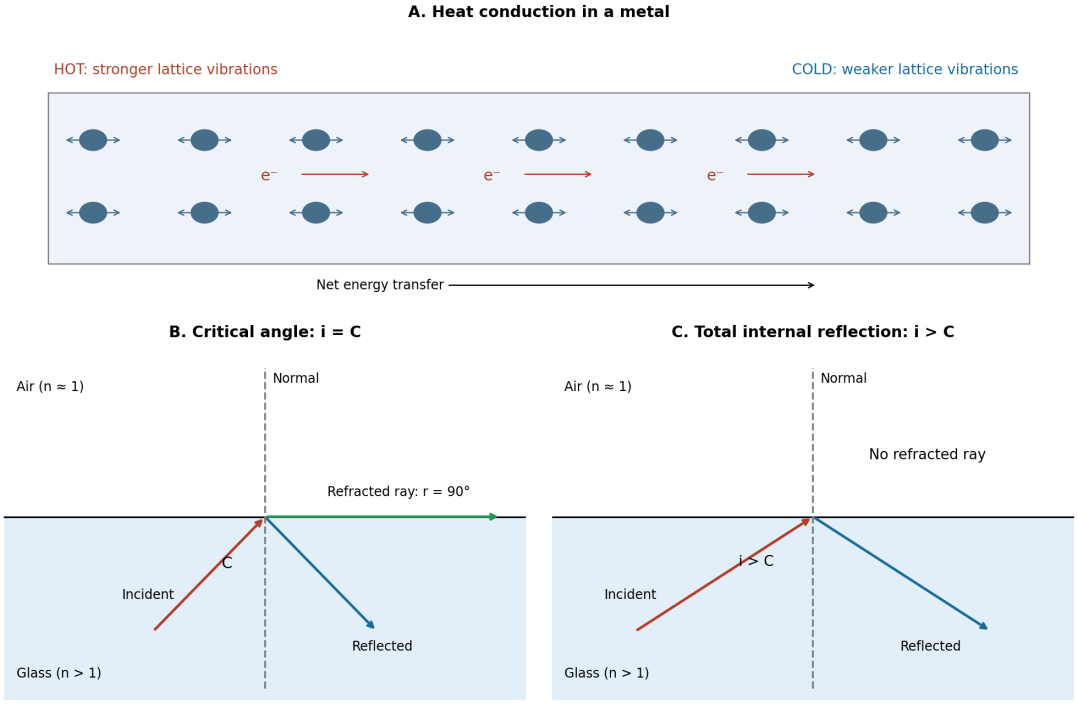
Given: P = 2000 W = 2 kW; t = 5 h; rate = Rs. 15/kWh
Formula: E = Pt; cost = E × rate
Working: E = 2 × 5 = 10 kWh; cost = 10 × 15
Answer: Cost = Rs. 150; energy = 10 units.

Q.5 Explain critical angle and total internal reflection. State the necessary conditions.

Critical angle C: the angle of incidence in an optically denser medium for which the refracted ray in the rarer medium makes 90° with the normal and travels along the interface.

Total internal reflection: complete reflection back into the denser medium. Both conditions are necessary: (1) light travels from higher refractive index to lower refractive index; (2) **i > C**. At i = C, the refracted ray is along the boundary; total internal reflection begins only above C.

Derivation: Snell's law gives $n_1 \sin i = n_2 \sin r$. At i = C, r = 90°, so $n_1 \sin C = n_2$. Hence **$\sin C = n_2/n_1$**. For a material of index n into air (index approximately 1), n sin C = 1, therefore **$n = 1/\sin C$**. All angles are measured from the normal. Optical fibres use repeated total internal reflection.



Panels B and C show the critical-angle and total-internal-reflection cases. Panel A concerns conduction.

Q.5 OR Differentiate analogue and digital electronics. Why is a digital system more reliable?

Analogue electronics processes continuously varying signals that can take any value within a range, as in an analogue audio amplifier. **Digital electronics** processes discrete signal levels, usually binary 0 and 1, as in a digital calculator.

Greater reliability:
1. Small amounts of noise usually do not change a digital voltage's recognised logic value because a range of voltages represents each state.
2. Digital signals can be regenerated to their intended levels and shape, limiting accumulated distortion.
3. Extra bits allow error detection and, in suitable systems, correction. Digital information can therefore be stored and transmitted consistently within the system's operating limits.

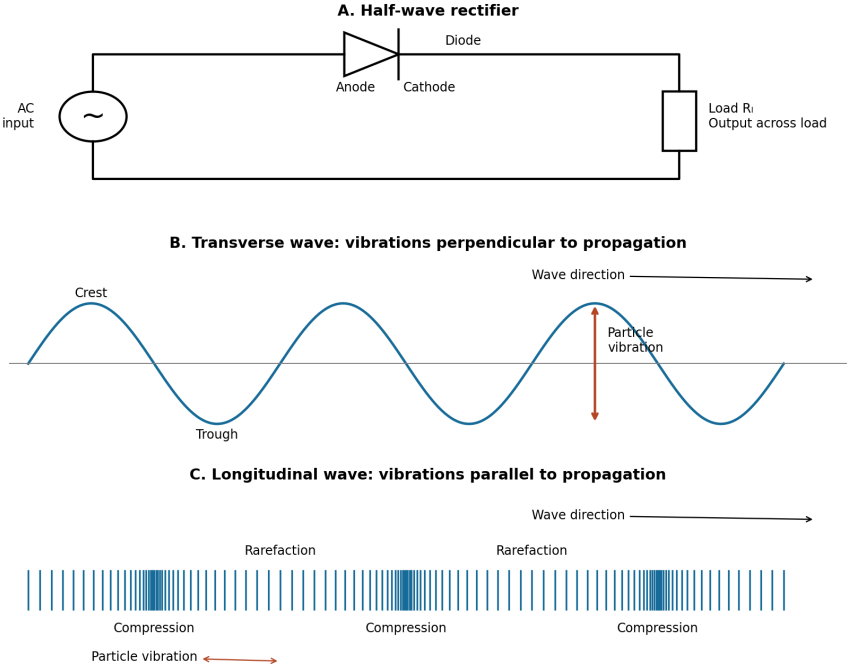
Q.6 Describe a diode under forward and reverse bias.

A **diode** is a p–n junction device that conducts readily in one direction and strongly opposes current in the other. A depletion region and potential barrier form at the junction.

Forward bias: connect p/anode towards positive and n/cathode towards negative. The applied voltage reduces the barrier and narrows the depletion layer. Majority carriers cross more readily and appreciable current flows when the voltage is sufficient. The conducting diode has low effective resistance.

Reverse bias: connect p towards negative and n towards positive. The barrier rises and the depletion region widens. Majority-carrier flow is blocked; only a tiny leakage current normally flows. The resistance is high. Excessive reverse voltage causes breakdown and potentially a large current.

This one-direction behaviour allows **rectification**: converting a.c. into unidirectional current.



Panel A gives a diode rectifier circuit; the diode conducts on its forward-biased half-cycle. Reverse the diode polarity relative to the supply for reverse bias.

Q.6 OR Describe nuclear fusion with an example and explain energy production in stars such as the Sun.

Nuclear fusion combines light nuclei into a heavier nucleus and releases energy. Example: **$^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n + 17.6$ MeV.**

Energy in stars: In the Sun, a sequence of reactions (mainly the proton–proton chain) converts hydrogen to helium. The extremely hot core gives nuclei high kinetic energy, allowing close encounters; quantum tunnelling allows some nuclei to fuse despite their electrostatic repulsion. High density sustained by gravity permits continuing reactions.

The products have a smaller total rest mass; the difference releases energy according to **$E = \Delta mc^2$**. A school-level net equation is **$4\ {}^{1}_{1}H \rightarrow {}^{4}_{2}He$ + 2 positrons + 2 electron neutrinos + energy**. Energy eventually leaves the surface as radiation. The deuterium–tritium example is a fusion reaction, not the Sun's main reaction route.

Paper 2 | 1st Annual 2026 - Set 1

Identified by MCQ 1: “The main function of a diode in a circuit is to…”, and Q.2(i): isotopes / heat calculation. All supplied MCQs, short and long questions are included.

Section A | MCQs with reasoning

1. The main function of a diode in a circuit is to:

Correct option: D - Rectify current
It conducts mainly in one direction, enabling conversion of a.c. to pulsating d.c.

2. The critical angle for polystyrene (n = 1.49) is:

Correct option: D - 42.2°
For polystyrene to air, $C = \sin^{-1}(1/1.49) \approx 42.2°$.

3. Charging solids by friction occurs due to transfer of:

Correct option: D - Electrons
Electrons move between materials; protons remain bound in nuclei.

4. The output of an OR gate is LOW when:

Correct option: C - All inputs are LOW
Any HIGH input makes OR output HIGH; only all-zero inputs give zero.

5. Warm air rises and cool air sinks in a heated room because of:

Correct option: C - Convection
Warm air expands and becomes less dense; cooler denser air sinks, producing circulation.

6. If current direction is reversed, the magnetic force on a conductor will:

Correct option: A - Reverse
With the magnetic field unchanged, reversing current reverses force (Fleming’s left-hand rule).

7. A charge of 10 C flows in 5 s. Find current.

Correct option: C - 2 A
$I = Q/t = 10/5 = 2$ A.

8. Linear expansion in solids refers to a change in:

Correct option: D - Length only
Linear expansion describes the length dimension, rather than area or volume.

9. Iodine has an 8-day half-life. How much of 40 g remains after 16 days?

Correct option: C - 10 g
Two half-lives: 40 → 20 → 10 g.

10. Compression and rarefaction are features of which waves?

Correct option: A - Longitudinal
Particles oscillate parallel to propagation, creating alternating dense and sparse regions.

11. Which instrument converts electrical energy into mechanical energy?

Correct option: B - Electric motor
A motor uses magnetic forces on current-carrying conductors to produce motion.

12. Sound travels faster in solids than gases mainly because solids have:

Correct option: No scientifically correct option; B appears intended
Options are less pressure, higher density, lower density and fewer particles. Higher stiffness/elastic modulus is the main missing reason. Higher density alone does not make sound faster: for a thin rod $v = \sqrt{(Y/\rho)}$. B ("higher density") may be the examiner's intended answer, but should not be taught as the physical cause. An official key was not supplied.

Section B | Q.2: short answers

Q.2 (i) What is an isotope? Do isotopes have similar chemical properties?

Isotopes are atoms of the same element with the **same atomic number but different mass numbers**: equal proton numbers but different neutron numbers. Example: carbon-12 and carbon-14. They have similar chemical properties because neutral atoms of the isotopes have the **same electronic configuration and valence-electron number**.

OR Calculate heat to raise 2 kg of a substance from 40 °C to 45 °C; c = 420 J kg^{-1} K^{-1}.

Given: m = 2 kg; c = 420 J kg^{-1} K^{-1}; ΔT = 5 K
Formula: Q = mcΔT
Working: Q = 2 × 420 × 5
Answer: Q = 4200 J = 4.2 kJ.

Q.2 (ii) What is an electric field? How does it affect a test charge?

An electric field is the region in which a charge experiences an electric force. Field strength is force per unit positive test charge: **E = F/q**. A charge experiences **F = qE**. A positive charge experiences force along the field, and a negative charge opposite to it. A free charge accelerates if the resultant force is unbalanced.

OR State two functions of a diode in electronic circuits.

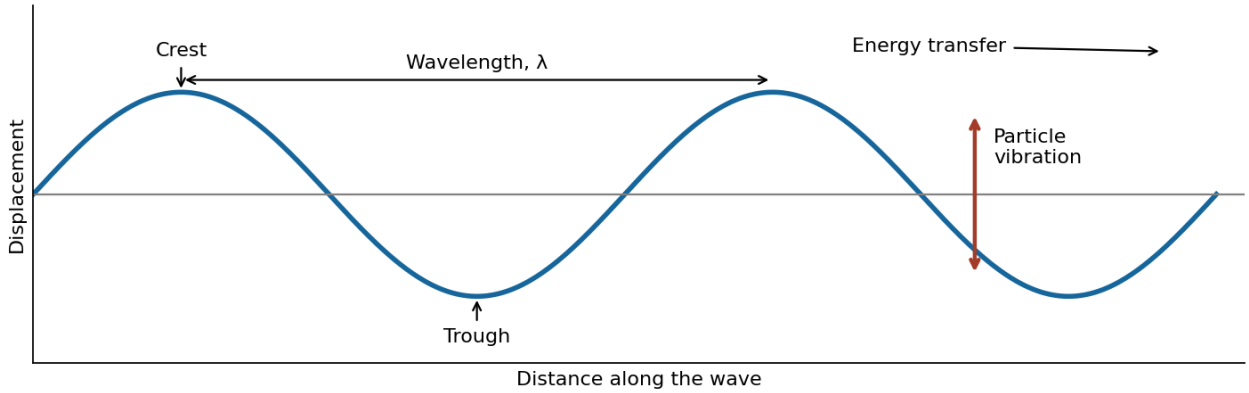
1. Rectification: a diode conducts mainly in one direction and converts a.c. into unidirectional, pulsating d.c.
2. Switching: it acts as a conducting switch under sufficient forward bias and a non-conducting switch under reverse bias below breakdown.

Q.2 (iii) Draw a transverse waveform. How is energy transferred in a transverse wave?

In a transverse wave, medium particles vibrate **perpendicular** to the direction of energy transfer. For a wave travelling horizontally on a rope, particles move up and down about equilibrium while energy travels horizontally. The particles do not travel with the wave, so there is no net transfer of matter.

OR Find total e.m.f. of three 1.5 V cells in series and parallel.

Series, same sense: E = 1.5 + 1.5 + 1.5 = **4.5 V**.
Parallel, identical cells with like terminals connected: E equals that of one cell = **1.5 V**.



Particle vibration is perpendicular to the direction of energy transfer.

Q.2 (iv) Find cost of using a 1000 W heater for 5 hours at Rs. 30 per unit.

Given: P = 1000 W = 1 kW; t = 5 h
Formula: E = Pt; cost = E × rate
Working: E = 1 × 5 = 5 kWh; cost = 5 × 30
Answer: Rs. 150.

OR Define refractive index and calculate it when light speed is 2 × 10^8 m/s.

Absolute refractive index is the ratio of light speed in vacuum to its speed in the medium: **n = c/v**. It has no unit. Here, **n = (3 × 10^8)/(2 × 10^8) = 1.5**.

Q.2 (v) State Lenz's law. How does it determine induced-current direction?

The induced current flows so that its magnetic field **opposes the change in magnetic flux producing it**. If a magnet's north pole approaches a coil, the near face becomes north, opposing the approach. Viewed from the approaching magnet, current is anticlockwise; the right-hand grip rule gives the direction.

OR Find power of a 220 V appliance drawing 2 A.

Given: V = 220 V; I = 2 A
Formula: P = VI
Working: P = 220 × 2
Answer: P = 440 W.

Q.2 (vi) State two differences between d.c. and a.c.

D.c.: current flows in one direction; steady d.c. has constant magnitude. A cell supplies d.c.
A.c.: current reverses direction periodically and its instantaneous value varies periodically. The mains supply provides a.c.

OR Why do coastal areas have more moderate temperatures than inland areas?

Seawater has a higher specific heat capacity than land, so its temperature changes more slowly. During the day or summer it absorbs much heat with a small temperature rise, keeping coastal areas cooler. At night or in winter it releases stored heat slowly, keeping them warmer. Air circulation distributes heat, reducing daily and seasonal temperature extremes.

Q.2 (vii) What is an echo? How is it produced?

An echo is a repetition of sound heard separately after reflection from a distant hard surface. Sound travels to the surface and back. A distinct echo needs approximately **0.1 s** delay. With v = 340 m/s, **d = vt/2 = 340 × 0.1/2 = 17 m** for source and listener close together.

OR Describe moisture's role in discharging an insulator.

Moisture forms a thin water film on an insulator. Dissolved impurities supply mobile ions, making the surface slightly conducting. Surface resistance falls, allowing accumulated charge to leak to surroundings or Earth through available paths. Static charge is therefore lost more rapidly in humid conditions.

Q.2 (viii) Name six electromagnetic regions in increasing wavelength.

Six regions in **increasing wavelength** are:
Gamma rays → X-rays → Ultraviolet → Visible light → Infrared → Microwaves.
Radio waves follow for the full spectrum. Frequency decreases as wavelength increases: c = fλ.

OR State Ohm's law and find p.d. for a 10 Ω resistor carrying 2 A.

At constant temperature and other physical conditions, current through a conductor is directly proportional to the potential difference across it: **V ∝ I; V = IR**.
Given: R = 10 Ω; I = 2 A.
V = IR = 2 × 10 = 20 V.

Q.2 (ix) Define half-life with an example.

Half-life is the time required for half the undecayed radioactive nuclei in a sample to decay; its activity also falls to half.
$\mathbf{T_{½} = 0.693/\lambda}$, where λ is the decay constant. The remaining quantity is $\mathbf{N = N_0(½)^{t/T½}}$.
Example: with an 8-day half-life, 40 g of radioactive iodine becomes 20 g after 8 days and 10 g after 16 days.

OR An object 2 cm from a convex lens forms an image at 6 cm. Find focal length and magnification.

Assumption: a real image lies on the opposite side. Cartesian signs: u = −2 cm, v = +6 cm.
$\mathbf{1/f = 1/v - 1/u = 1/6 + 1/2 = 2/3\ cm^{-1}}$, so **f = 1.5 cm**.
m = v/u = 6/(−2) = −3: real, inverted, three times object height.
Ambiguity: image type is not stated. If the image is virtual, v = −6 cm gives f = 3 cm and m = +3. State the real-image assumption in the exam.

Q.2 (x) Give three differences between real and virtual images.

Formation: real images form where rays actually meet; virtual images form where rays appear to meet when extended backwards.
Screen: a real image can be obtained on a screen at its position; a virtual image cannot be directly projected there.
Orientation: for a single lens or mirror and a real object, real images are inverted and virtual images upright.

OR How does wavelength affect diffraction through a gap?

For a fixed gap width, **longer wavelength gives greater diffraction** and shorter wavelength gives less spreading. Diffraction is especially noticeable when wavelength is comparable to or greater than the gap width. The relevant comparison is wavelength relative to opening width.

Q.2 (xi) State three differences between boiling and evaporation.

Temperature: boiling occurs at a definite boiling point for a given pressure; evaporation can occur at any temperature at which the substance is liquid.
Location: boiling occurs throughout the liquid; evaporation occurs at its surface.
Bubbles: boiling forms vapour bubbles within the liquid; evaporation occurs without such bubbles.

OR What is the clear-day sky colour? Explain using scattering.

The clear daytime sky appears **blue**. Air molecules scatter shorter visible wavelengths much more strongly than longer wavelengths, so blue light reaches us from many directions. For Rayleigh scattering, **scattering intensity $\propto 1/\lambda^4$**. Violet is also strongly scattered, but the solar spectrum and the eye's response make the sky appear predominantly blue.

Section C | Long answers

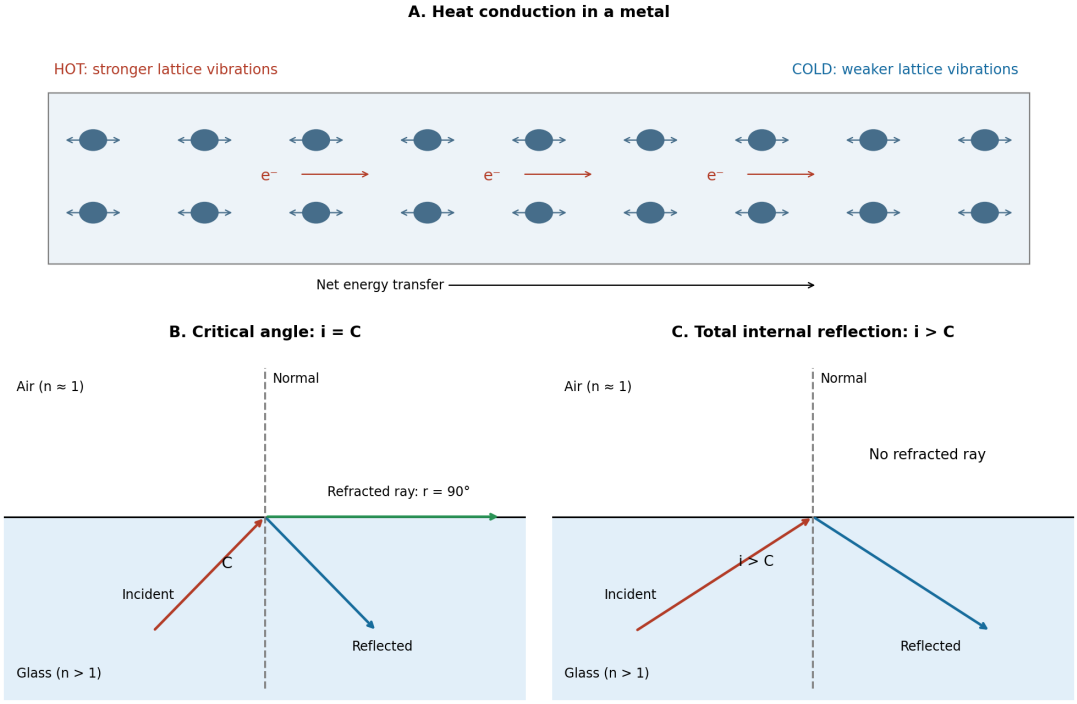
Q.3 Explain conduction through lattice vibrations and free electrons, with diagram.

Conduction transfers thermal energy from hot to cold without bulk movement of the material.

Lattice vibrations: Solid particles vibrate about fixed equilibrium positions. Heating increases vibrational energy at the hot end. Interactions with neighbouring particles transfer energy along the lattice while particles remain near their equilibrium positions.

Free electrons: Metals contain mobile electrons. Electrons in the hotter region gain energy and transfer it through movement and collisions/interactions with electrons and lattice ions. Their random motion produces a net energy transfer towards the colder region.

Comparison: Metals conduct by both electrons and lattice vibrations and are generally good thermal conductors (copper, aluminium). Non-metallic solids generally lack mobile electrons, conduct mainly through lattice vibrations, and are often poorer conductors (wood, plastic). Exceptions exist, such as diamond. The electron arrows in the diagram show net energy transport, not all electrons moving in one direction.



Panel A: energy travels from the hot region to the cold region by lattice interactions and electrons.

Q.3 OR (a) Describe fusion and why high temperature is needed. (b) Find energy when 55.56 kg is converted into energy.

(a) Fusion: two light nuclei combine into a heavier nucleus and release energy, e.g. $^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n$ **+ energy**. Very high temperature gives nuclei high kinetic energy, enabling them to approach closely enough for the short-range strong nuclear interaction to bind them, despite electrostatic repulsion.

(b) Given: $\Delta m = 55.56$ kg; $c = 3.0 \times 10^{8}$ m/s.
Formula: $E = \Delta mc^{2}$.
Substitution: $E = 55.56 \times (3.0 \times 10^{8})^{2} = 55.56 \times 9 \times 10^{16}$ J.
Answer: $E = 5.0004 \times 10^{18}$ J $\approx 5.0 \times 10^{18}$ J.

Q.4 (a) Describe factors affecting loudness and pitch. (b) Explain amplitude's effect on loudness, with examples.

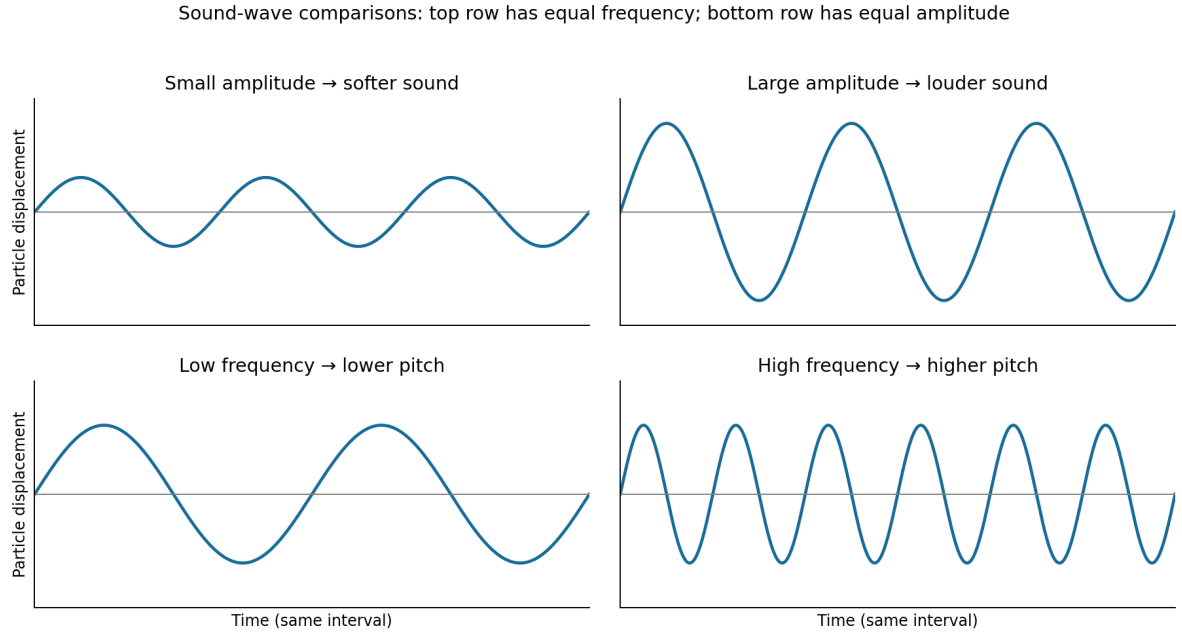
Loudness distinguishes loud from soft sound and depends mainly on intensity reaching the ear. For the same medium and frequency, $\mathbf{I \propto A^{2}}$. Increasing amplitude makes sound generally louder; reducing it makes sound softer. Distance and the ear's sensitivity also influence perceived loudness.

Pitch distinguishes high/shrill from low/deep sound and depends mainly on frequency. More vibrations per second produce a higher pitch; fewer produce a lower pitch. Thus a 1000 Hz tone has a higher pitch than a 200 Hz tone.

Examples: Hit a drum harder or pluck a string more strongly: amplitude and loudness increase, while pitch remains approximately unchanged. Tighten the same string: vibration frequency and pitch increase.

If amplitude doubles, intensity becomes four times as large: $\mathbf{I_2/I_1 = (2A/A)^{2} = 4}$. Perceived loudness does not necessarily become four times greater. In the diagrams, the top pair keeps frequency constant; the bottom pair keeps amplitude constant. These are displacement-time

plots, not drawings of transverse sound propagation in air.



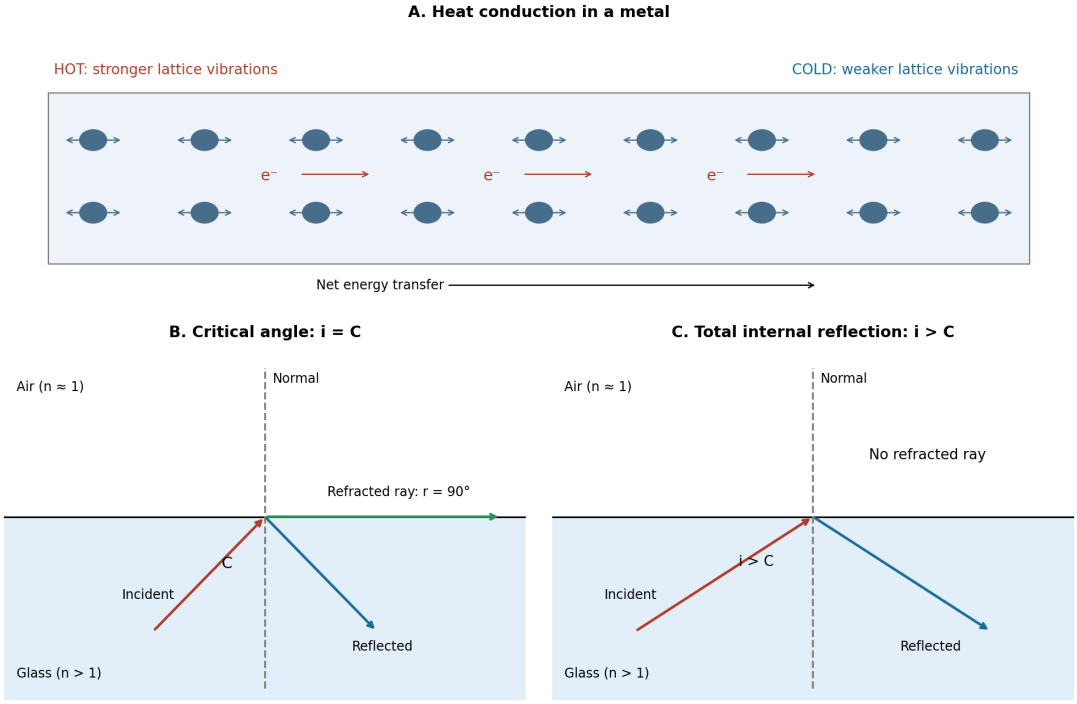
Top: loudness comparison at equal frequency. Bottom: pitch comparison at equal amplitude.

Q.4 OR (a) Explain total internal reflection with a ray diagram. (b) Derive n = 1/sin C.

Critical angle C: the angle of incidence in an optically denser medium for which the refracted ray in the rarer medium makes 90° with the normal and travels along the interface.

Total internal reflection: complete reflection back into the denser medium. Both conditions are necessary: (1) light travels from higher refractive index to lower refractive index; (2) **i > C**. At i = C, the refracted ray is along the boundary; total internal reflection begins only above C.

Derivation: Snell's law gives $n_1 \sin i = n_2 \sin r$. At i = C, r = 90°, so $n_1 \sin C = n_2$. Hence **$\sin C = n_2/n_1$**. For a material of index n into air (index approximately 1), n sin C = 1, therefore **$n = 1/\sin C$**. All angles are measured from the normal. Optical fibres use repeated total internal reflection.



Use panels B and C. At C the transmitted ray is along the boundary; above C there is total internal reflection.

Q.5 (a) Define linear expansion. (b) Explain why gaps are left in railway tracks.

(a) Linear expansion: the increase in a solid's length with a rise in temperature. Particle vibrations increase and their average separation generally increases.
$\Delta L = \alpha L_0 \Delta T$, where L_0 is original length, α is the coefficient of linear expansion, and ΔT is the temperature rise. The new length is $L = L_0(1 + \alpha\Delta T)$.

(b) Railway gaps: Metal rails expand in hot weather. Gaps in jointed tracks provide room for this expansion. Without allowance, adjacent rails push against each other, creating large compressive forces that may buckle the track and cause accidents. Modern continuously welded tracks use engineered stress management rather than ordinary gaps at every rail length.

Q.5 OR Derive resistance of two parallel resistors and calculate it for 6 Ω and 3 Ω.

Connect R_1 and R_2 across the same two nodes. The potential difference is the same across each: **$V_1 = V_2 = V$**. Total current divides: **$I = I_1 + I_2$**.

By Ohm's law, $I = V/R_p$, $I_1 = V/R_1$ and $I_2 = V/R_2$. Substitution gives:
$V/R_p = V/R_1 + V/R_2$.
Dividing by V gives **$1/R_p = 1/R_1 + 1/R_2$**.
Taking a common denominator:
$1/R_p = (R_1 + R_2)/(R_1R_2)$.
Therefore **$R_p = R_1R_2/(R_1 + R_2)$**.

The equivalent resistance is smaller than either positive individual resistance.

Given: $R_1 = 6\ \Omega$; $R_2 = 3\ \Omega$
Formula: $R_p = R_1R_2/(R_1 + R_2)$
Working: $R_p = 6 \times 3/(6 + 3) = 18/9$
Answer: $R_p = 2\ \Omega$.

Q.6 State the transformer principle, explain working and formula, and describe construction with diagram.

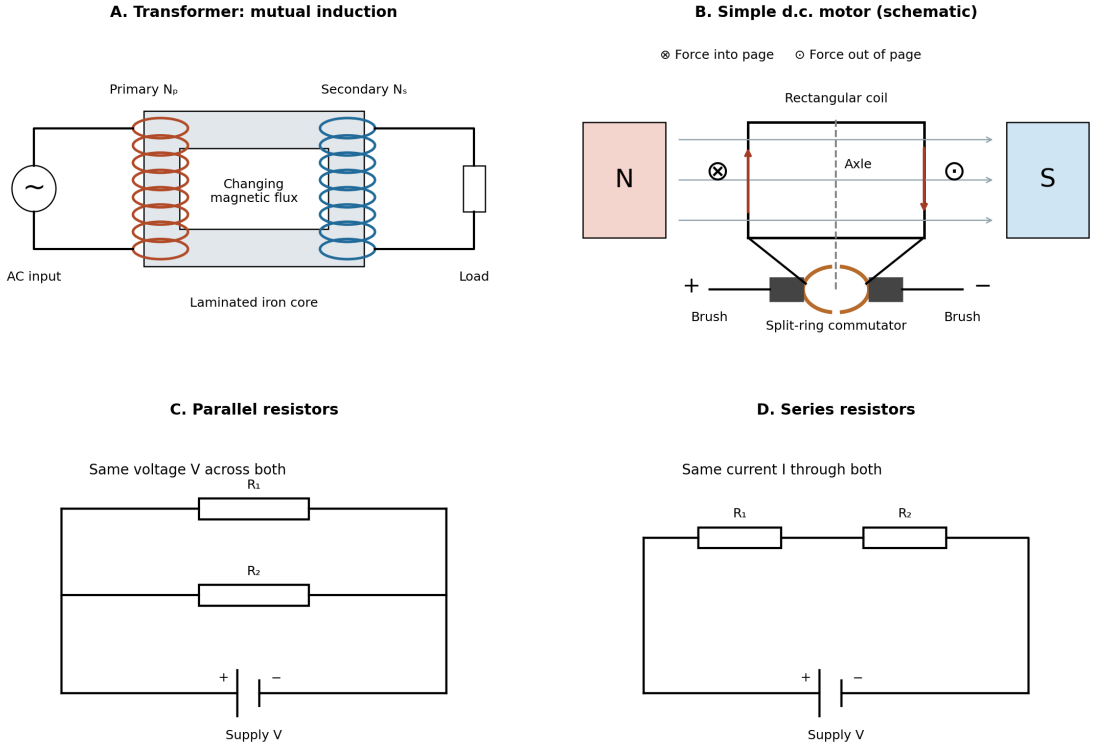
Principle: mutual electromagnetic induction. A changing current in the primary creates changing magnetic flux that induces e.m.f. in the secondary.

Construction: Two insulated copper-wire coils are wound on a closed laminated soft-iron core. The primary connects to the a.c. input and the secondary to the load. Coils are electrically insulated from each other and the core. The core guides flux; its laminations reduce eddy-current losses.

Working: Alternating current in the primary creates alternating magnetic flux in the core. Flux links the secondary, inducing an alternating e.m.f. When a load closes the secondary circuit, current flows. A steady d.c. supply cannot sustain transformer action.

Turns relation: $V_s/V_p = N_s/N_p$ (ideal transformer). $N_s > N_p$ gives a step-up transformer; $N_s < N_p$ gives a step-down transformer. At equal ideal power, $\mathbf{V_pI_p = V_sI_s}$, so increasing voltage reduces current.

Energy transfer: electrical input → changing magnetic field → electrical output. The transformer changes voltage/current levels; it does not create energy. Real devices lose some energy as heat in the windings and core.



Panel A: transformer construction. Panel C also illustrates the parallel connection in Q.5 OR.

Q.6 OR Explain one-direction current conduction and light emission by an LED; draw its diagram.

An **LED (light-emitting diode)** is a semiconductor p–n junction that emits light when an appropriate forward current flows.

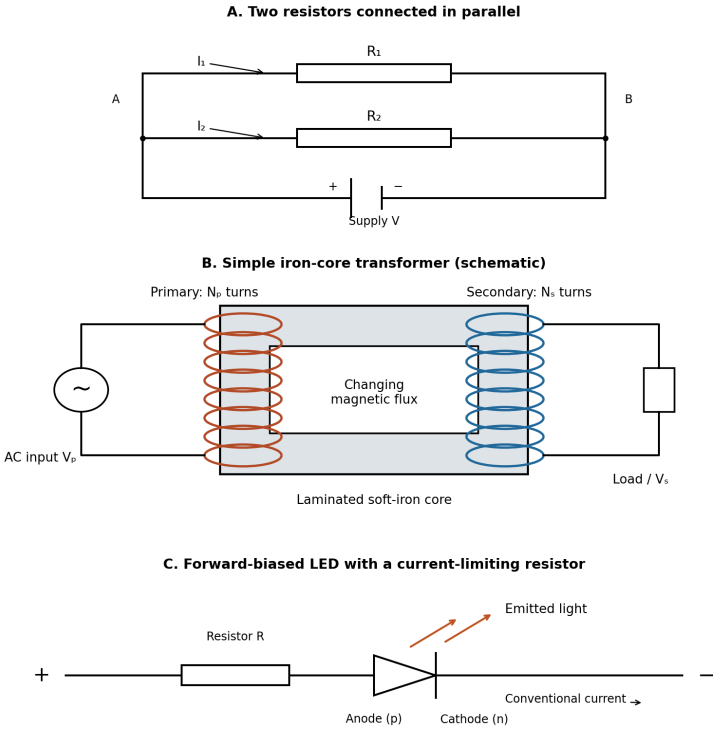
Forward conduction: connect p-side/anode towards positive and n-side/cathode towards negative. The barrier falls and depletion layer narrows. With sufficient voltage, electrons and holes

cross the junction and recombine.

Light emission: Recombination in the LED material releases energy as photons. The semiconductor energy gap determines colour. Electrical energy becomes light and heat.

Reverse bias: the barrier and depletion width increase, blocking appreciable current below breakdown. The LED does not emit useful light; excessive reverse voltage can damage it.

Series resistor: limits current to protect the LED. $R = (V_{supply} - V_{LED})/I_{LED}$. The two outward arrows on the LED symbol represent light emission.



Use panel C: LED forward bias, current-limiting resistor and outward light arrows.

Paper 3 | 1st Annual 2026 - Set 2

Identified by MCQ 1: “A diode conducts electricity when it is…”, and Q.2(i): alpha particles / specific heat. Section B (ix)–(xi) and their OR options are missing from the uploads; they are not reconstructed.

Section A | MCQs with reasoning

1. A diode conducts electricity when it is:

Correct option: C - Forward biased
Forward bias lowers the barrier and permits appreciable current when sufficient voltage is applied.

2. Speed of light in glass with n = 1.5 is:

Correct option: B - 2×10^8 m/s
$v = c/n = (3 \times 10^8)/1.5 = 2 \times 10^8$ m/s.

3. Which charge is transferred during charging by friction?

Correct option: A - Negative charge
Electrons carry negative charge and move between the materials.

4. For a NOR gate, the expression is:

Correct option: C - X = NOT(A + B)
NOR is an OR gate followed by inversion; only A = B = 0 gives output 1.

5. A metal cooking pan heats quickly mainly due to:

Correct option: D - Conduction
Energy spreads through metal by lattice vibrations and mobile electrons.

6. Reversing magnetic field while keeping current constant makes force:

Correct option: A - Reverse direction
The force direction depends on both current and magnetic field; reversing one reverses force.

7. If charge passing in a given time doubles, current will:

Correct option: C - Double
I = Q/t. At fixed t, doubling Q doubles I.

8. Which generally expands most when heated?

Correct option: D - Gases
For the same temperature rise, gases generally show the greatest fractional volume expansion at constant pressure.

9. Which radiation is stopped by thin paper?

Correct option: D - Alpha particles
Alpha particles have low penetrating power and short range.

10. Wave speed is given by:

Correct option: B - $v = f\lambda$
Frequency × wavelength has unit $s^{-1} \times m = m/s$.

11. Which motor component reverses current?

Correct option: D - Split-ring commutator
It reverses coil current each half-turn to keep torque in the same rotational sense.

12. In which medium does sound generally travel fastest?

Correct option: D - Solids
High stiffness generally permits fast sound propagation. Speed depends on elasticity and density together; sound cannot propagate in vacuum.

Section B | Available short answers

Q.2 (i) Describe alpha particles and give a reaction example.

An alpha particle is a helium nucleus, **$^{4}_{2}He$**, containing two protons and two neutrons. It has charge **+2e** and mass approximately **4 u**. It is strongly ionising but weakly penetrating; paper can stop it.
Example: $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$ + energy.

OR Define specific heat capacity and give its mathematical expression.

Specific heat capacity is the energy required to raise the temperature of **1 kg** of a substance by **1 K or 1 °C**, without a change of state.
$c = Q/(m\Delta T)$, or **$Q = mc\Delta T$**.
SI unit: **$J\ kg^{-1}\ K^{-1}$**.

Q.2 (ii) Define electric field and state the effect on a positive test charge.

An electric field is the region in which a charge experiences an electric force. Field strength is force per unit positive test charge: **$E = F/q$**. A charge experiences **$F = qE$**. A positive charge experiences force along the field, and a negative charge opposite to it. A free charge accelerates if the resultant force is unbalanced.

OR What is the role of a diode in rectification? Explain with a diagram.

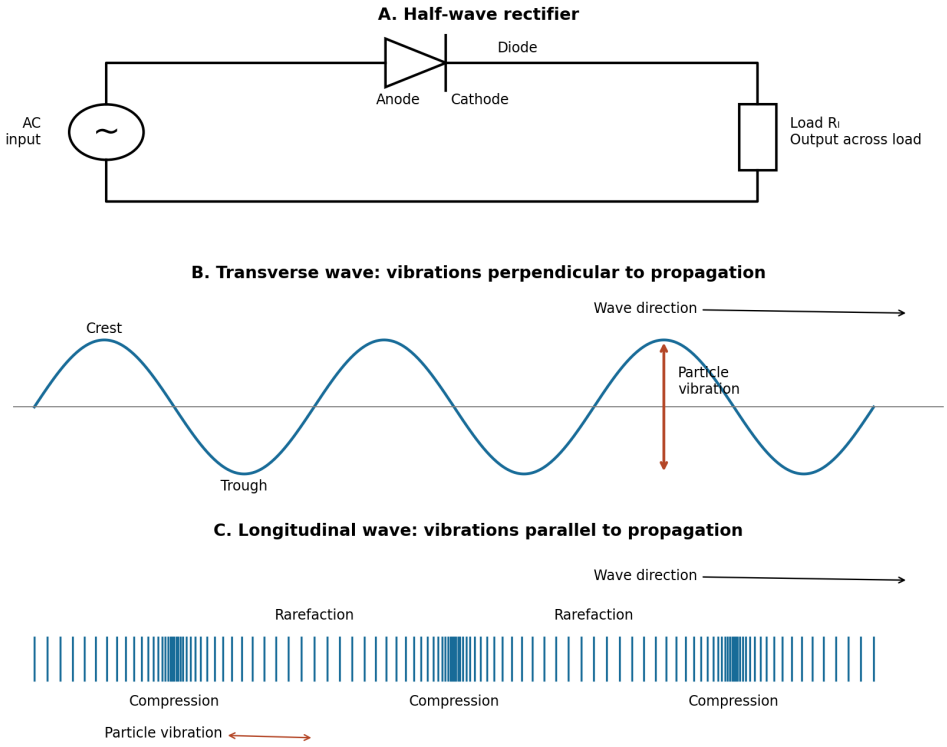
A diode rectifies because it conducts mainly in one direction. In a half-wave rectifier, one a.c. half-cycle forward biases it and current flows through the load. The opposite half-cycle reverse biases it and current is practically blocked. The load receives **unidirectional, pulsating d.c.**, not smooth d.c.

Q.2 (iii) Draw waveforms comparing vibration directions in transverse and longitudinal waves.

Transverse: particles vibrate perpendicular to propagation; the pattern has crests and troughs. Example: a stretched-string wave.
Longitudinal: particles vibrate parallel to propagation, forming compressions and rarefactions. Example: sound in air.
In both, particles oscillate locally while energy travels through the medium.

OR Find total e.m.f. of four 2 V cells in series and parallel.

Series (same sense): E = 4 × 2 = **8 V**.
Parallel (identical cells, like terminals together): E = **2 V**, the e.m.f. of one cell.



Panel A: half-wave rectifier. Panels B and C compare perpendicular and parallel particle vibration.

Q.2 (iv) A 100 W bulb operates for 10 hours daily. Calculate one day's energy.

Given: P = 100 W = 0.1 kW; t = 10 h
Formula: E = Pt
Working: E = 0.1 × 10 = 1 kWh; also E = 100 × (10 × 3600) J
Answer: E = 1 kWh = 3.6 × 10⁶ J = one electricity unit.

OR State the principle of optical fibres and two advantages.

Optical fibres work by **total internal reflection**. A higher-index core is surrounded by lower-index cladding. Light incident at the core-cladding boundary above the critical angle is repeatedly reflected along the core.
Advantages: high information capacity/data rates and low signal attenuation over long distances.

Q.2 (v) State Lenz's law. Why does induced current oppose the producing change?

Lenz's law states that the induced current's magnetic field opposes the change in magnetic flux producing it. When a north pole approaches a coil, its near face becomes north and repels the magnet. External mechanical work must be done against this force; this supplies the electrical energy produced. If induction assisted its own cause, energy could be created without equivalent input, violating conservation of energy.

OR Find bulb power for I = 0.5 A at 220 V.

Given: V = 220 V; I = 0.5 A
Formula: P = VI
Working: P = 220 × 0.5
Answer: P = 110 W.

Q.2 (vi) Why is a.c. preferred for domestic use? Give three reasons.

1. A.c. voltage can be changed efficiently using transformers.
2. Stepping up voltage reduces current for a given power, reducing transmission losses $\mathbf{P_{loss} = I^2R}$.
3. Near consumers it can be stepped down to the domestic voltage; appliances can further transform or rectify it as needed.

OR How do sea and land breezes moderate coastal climate?

Day: cooler air moves from sea to land as warmer air over land rises; this sea breeze reduces daytime heating.
Night: cooler air moves from land to sea as air above the relatively warmer sea rises. This circulation redistributes heat. Along with seawater's high specific heat capacity, these breezes help moderate coastal temperature variations.

Q.2 (vii) State the minimum delay for an echo and explain its absence in a small room.

A distinct echo requires a delay of approximately **0.1 s**. For v = 340 m/s, the reflecting surface should be about **d = vt/2 = 17 m** away when source and listener are close. In a small room reflections return sooner, merge with the original sound, and do not form a distinct echo.

OR How can a charged insulator be discharged using a flame?

A flame produces positive ions and electrons, making nearby gas conducting. Oppositely charged particles move towards the charged surface and neutralise its charge: electrons or negative ions neutralise positive charge, while positive ions neutralise negative charge. Thus the insulator is discharged.

Q.2 (viii) List six electromagnetic regions from higher to lower wavelength.

Six regions in **decreasing wavelength** are:
Radio → Microwaves → Infrared → Visible → Ultraviolet → X-rays.
Gamma rays follow in the complete sequence. Frequency increases in this order.

OR State Ohm's law. A 12 V p.d. produces 3 A. Find resistance.

At constant temperature and other physical conditions, current through a conductor is directly proportional to the potential difference across it: **V ∝ I; V = IR**.
R = V/I = 12/3 = 4 Ω.

Source gap: Q.2 (ix), (x) and (xi), with their OR alternatives, were not present in this set's uploaded images. Please supply that missing strip/page to complete this section.

Section C | Long answers

Q.3 (a) Explain conduction in metals through free electrons. (b) Compare conduction in metals and non-metals.

Conduction transfers thermal energy from hot to cold without bulk movement of the material.

Lattice vibrations: Solid particles vibrate about fixed equilibrium positions. Heating increases vibrational energy at the hot end. Interactions with neighbouring particles transfer energy along the lattice while particles remain near their equilibrium positions.

Free electrons: Metals contain mobile electrons. Electrons in the hotter region gain energy and transfer it through movement and collisions/interactions with electrons and lattice ions. Their random motion produces a net energy transfer towards the colder region.

Comparison: Metals conduct by both electrons and lattice vibrations and are generally good thermal conductors (copper, aluminium). Non-metallic solids generally lack mobile electrons, conduct mainly through lattice vibrations, and are often poorer conductors (wood, plastic). Exceptions exist, such as diamond. The electron arrows in the diagram show net energy transport, not all electrons moving in one direction.

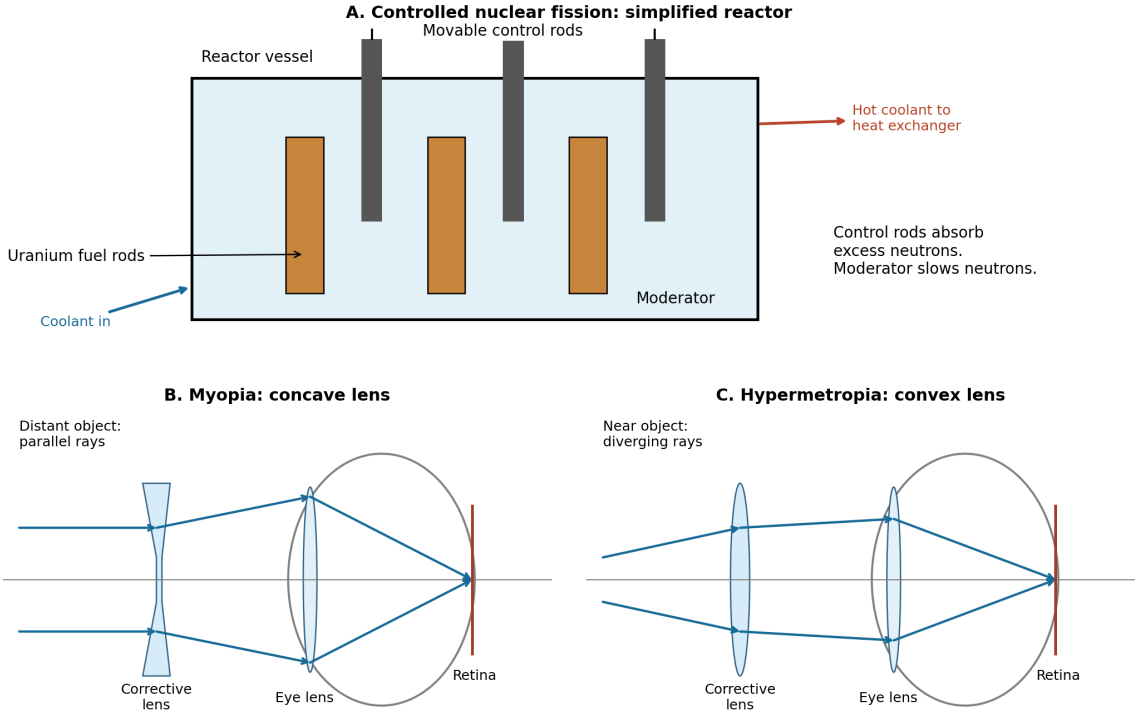
Q.3 OR (a) Describe the Sun's main energy source. (b) Explain controlled nuclear fission in reactors.

(a) Solar energy: The Sun is powered by nuclear fusion. In its hot, dense core, hydrogen is converted into helium through a sequence of reactions, mainly the proton–proton chain. The products have less total rest mass and the difference releases energy: $\mathbf{E = \Delta mc^2}$. Energy reaches the surface and leaves as radiation.

(b) Controlled fission: a heavy nucleus splits after absorbing a neutron, releasing lighter nuclei, energy and more neutrons. Example:

$$^{235}_{92}\mathbf{U} + {}^{1}_{0}\mathbf{n} \rightarrow {}^{141}_{56}\mathbf{Ba} + {}^{92}_{36}\mathbf{Kr} + 3\ {}^{1}_{0}\mathbf{n} + \textbf{energy.}$$

The released neutrons sustain a chain reaction. In a thermal reactor, a moderator (water or graphite) slows neutrons. Control rods containing boron or cadmium absorb excess neutrons; their position maintains on average one further fission per fission. Inserting rods reduces the rate; withdrawing them increases it. A coolant removes heat, which generates steam to drive a turbine and generator. The steady controlled reaction releases usable energy without a rapidly growing chain reaction.



Panel A: simplified thermal-reactor construction and control.

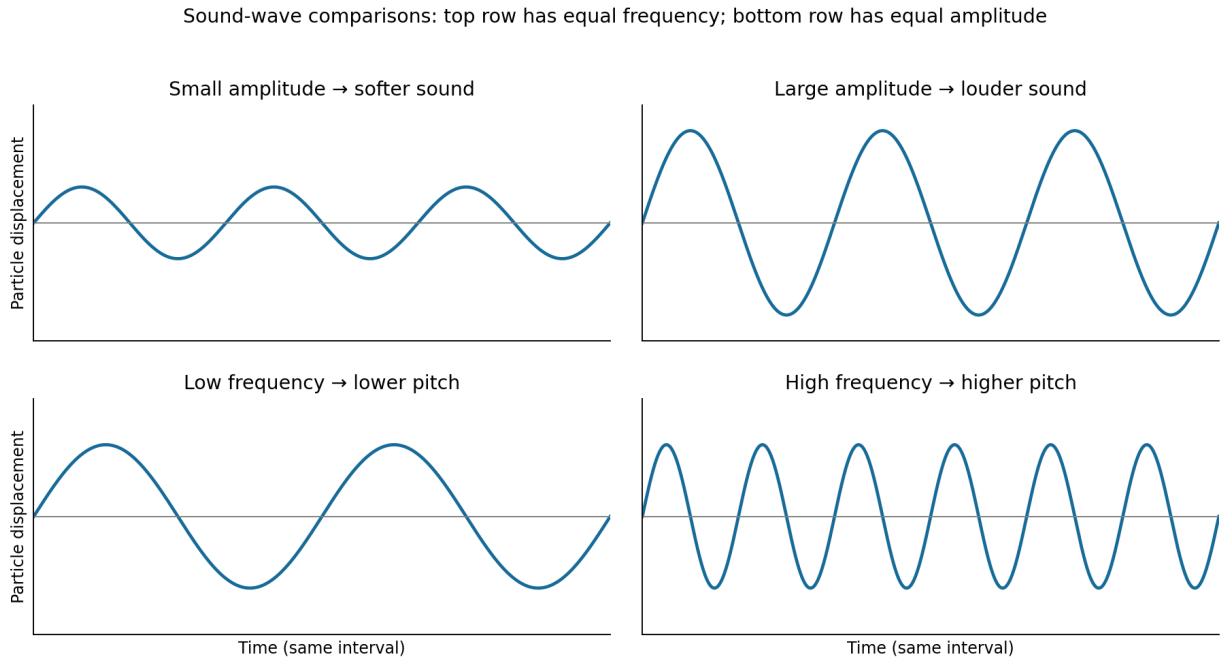
Q.4 (a) Explain effects of amplitude and frequency with diagrams. (b) Differentiate loudness and pitch with diagrams.

Loudness distinguishes loud from soft sound and depends mainly on intensity reaching the ear. For the same medium and frequency, $I \propto A^2$. Increasing amplitude makes sound generally louder; reducing it makes sound softer. Distance and the ear's sensitivity also influence perceived loudness.

Pitch distinguishes high/shrill from low/deep sound and depends mainly on frequency. More vibrations per second produce a higher pitch; fewer produce a lower pitch. Thus a 1000 Hz tone has a higher pitch than a 200 Hz tone.

Examples: Hit a drum harder or pluck a string more strongly: amplitude and loudness increase, while pitch remains approximately unchanged. Tighten the same string: vibration frequency and pitch increase.

If amplitude doubles, intensity becomes four times as large: $I_2/I_1 = (2A/A)^2 = 4$. Perceived loudness does not necessarily become four times greater. In the diagrams, the top pair keeps frequency constant; the bottom pair keeps amplitude constant. These are displacement-time plots, not drawings of transverse sound propagation in air.



Amplitude relates to loudness; frequency relates to pitch. Equal time intervals permit a valid frequency comparison.

Q.4 OR (a) Define critical angle and give its formula. (b) State conditions for total internal reflection.

Critical angle C: the angle of incidence in an optically denser medium for which the refracted ray in the rarer medium makes 90° with the normal and travels along the interface.

Total internal reflection: complete reflection back into the denser medium. Both conditions are necessary: (1) light travels from higher refractive index to lower refractive index; (2) **i > C**. At i = C, the refracted ray is along the boundary; total internal reflection begins only above C.

Derivation: Snell's law gives $n_1 \sin i = n_2 \sin r$. At i = C, r = 90°, so $n_1 \sin C = n_2$. Hence **$\sin C = n_2/n_1$**. For a material of index n into air (index approximately 1), n sin C = 1, therefore **$n = 1/\sin C$**. All angles are measured from the normal. Optical fibres use repeated total internal reflection.

Q.5 (a) Explain mutual induction in a transformer, with diagram. (b) Describe its energy transformation.

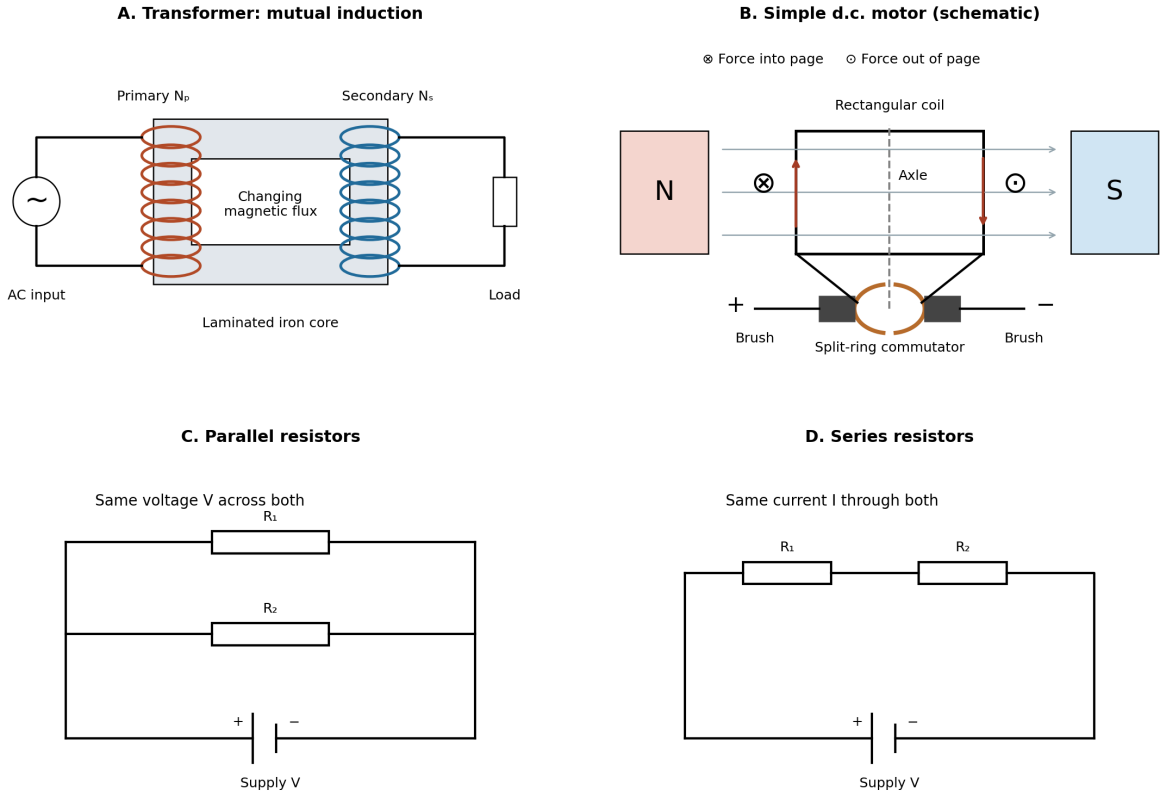
Principle: mutual electromagnetic induction. A changing current in the primary creates changing magnetic flux that induces e.m.f. in the secondary.

Construction: Two insulated copper-wire coils are wound on a closed laminated soft-iron core. The primary connects to the a.c. input and the secondary to the load. Coils are electrically insulated from each other and the core. The core guides flux; its laminations reduce eddy-current losses.

Working: Alternating current in the primary creates alternating magnetic flux in the core. Flux links the secondary, inducing an alternating e.m.f. When a load closes the secondary circuit, current flows. A steady d.c. supply cannot sustain transformer action.

Turns relation: $V_s/V_p = N_s/N_p$ (ideal transformer). $N_s > N_p$ gives a step-up transformer; $N_s < N_p$ gives a step-down transformer. At equal ideal power, **$V_pI_p = V_sI_s$**, so increasing voltage reduces current.

Energy transfer: electrical input → changing magnetic field → electrical output. The transformer changes voltage/current levels; it does not create energy. Real devices lose some energy as heat in the windings and core.



Panel A: transformer; panel B: motor; panels C and D: parallel and series circuits.

Q.5 OR (a) Derive the equivalent resistance of two resistors in parallel, with circuit diagram.

Connect R_1 and R_2 across the same two nodes. The potential difference is the same across each: $\mathbf{V_1 = V_2 = V}$. Total current divides: $\mathbf{I = I_1 + I_2}$.

By Ohm's law, $I = V/R_p$, $I_1 = V/R_1$ and $I_2 = V/R_2$. Substitution gives:
$\mathbf{V/R_p = V/R_1 + V/R_2.}$
Dividing by V gives $\mathbf{1/R_p = 1/R_1 + 1/R_2}$.
Taking a common denominator:
$1/R_p = (R_1 + R_2)/(R_1R_2)$.
Therefore $\mathbf{R_p = R_1R_2/(R_1 + R_2)}$.

The equivalent resistance is smaller than either positive individual resistance.

Q.5 OR (b) Derive the equivalent resistance of two resistors in series, with circuit diagram.

Connect R_1 and R_2 end to end. The same current passes through both: $\mathbf{I_1 = I_2 = I}$. The supply voltage equals the sum of the voltage drops: $\mathbf{V = V_1 + V_2}$.

Applying Ohm's law gives $\mathbf{IR_s = IR_1 + IR_2}$. Dividing by I:
$\mathbf{R_s = R_1 + R_2.}$
Thus the equivalent resistance of series resistors is the sum of their individual resistances.

Q.6 Describe an electric motor, split-ring commutator and brushes; draw a neat sketch.

Principle: A motor converts electrical energy into mechanical energy. A current-carrying conductor in a magnetic field experiences force; Fleming's left-hand rule gives its direction.

Construction: A rectangular insulated-wire coil rotates on an axle between magnetic poles. The coil ends connect to two insulated halves of a split-ring commutator. Stationary carbon brushes connect the commutator to a d.c. supply.

Working: Current flows in opposite directions in the two opposite sides of the coil. The field exerts opposite forces on them, forming a couple that turns the coil. After each half-turn, the split-ring commutator reverses the coil current. The forces reverse at the appropriate instant, keeping the turning effect in the same rotational sense. Momentum carries the coil through positions of zero turning effect.

Commutator: reverses coil current every half-turn to maintain rotation in one direction.
Brushes: maintain sliding electrical contact between stationary supply and rotating commutator. They do not themselves reverse the current.

Q.6 OR Make truth tables for AND, OR, NAND, NOR and NOT gates.

The requested tables are given below. Inputs and outputs are binary logic values.

For binary logic, 0 means LOW and 1 means HIGH. AND gives 1 only for two HIGH inputs; OR gives 1 if either is HIGH; NAND and NOR invert AND and OR respectively.

A	B	AND: A·B	OR: A+B	NAND: NOT(A·B)	NOR: NOT(A+B)
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

NOT has one input and reverses its value: X = NOT A.

Input A	NOT output X
0	1
1	0

Paper 4 | 1st Annual 2026 - Set 3

Identified by MCQ 1: “A diode is commonly used as...”, and Q.2(i): alpha-scattering evidence / heat needed for 25 kg of water. All supplied questions and OR alternatives are included.

Section A | MCQs with reasoning

1. A diode is commonly used as a:

Correct option: C - Rectifier
Its one-direction conduction enables rectification of a.c.

2. If incidence angle is zero, the refracted ray will:

Correct option: A - Continue along the normal
Snell’s law gives r = 0°. The ray does not bend, although speed/wavelength may change.

3. An object becomes negatively charged when it:

Correct option: D - Gains electrons
Gaining negatively charged electrons creates an excess of negative charge.

4. If both inputs of a NAND gate are HIGH, output is:

Correct option: D - LOW
NOT(1 AND 1) = NOT 1 = 0.

5. Which feature reduces heat loss from buildings?

Correct option: C - Thermal insulation
Insulation reduces thermal energy transfer through the building fabric.

6. Magnetic force on a current-carrying conductor is maximum when it is:

Correct option: D - Perpendicular to the field
$F = BIL \sin\theta$. At $\theta = 90°$, $\sin\theta = 1$ and force is maximum.

7. The correct formula for electric current is:

Correct option: A - $I = Q/t$
Current is charge passing a cross-section per unit time.

8. The coefficient of volume expansion β is:

Correct option: B - $\Delta V/(V_0 \Delta T)$
It is fractional increase in volume per unit temperature rise; unit K^{-1}.

9. Maximum displacement of a vibrating particle refers to:

Correct option: B - Amplitude
Amplitude is measured from equilibrium.

10. Wave speed is 7 m/s. If amplitude doubles, speed will:

Correct option: A - Not change
Speed stays 7 m/s in ordinary linear-wave conditions with the medium unchanged.

11. If the split-ring commutator is removed, the motor will:

Correct option: C - Stop after half rotation (intended)
Without current reversal, sustained unidirectional torque is lost. C is the expected school answer, assuming the coil remains connected. Literally removing the commutator may break the circuit; without reversal the coil may overshoot and oscillate rather than stop at exactly half a turn.

12. Sound cannot travel through:

Correct option: A - Vacuum
Sound requires a material medium to transmit mechanical vibrations.

Section B | Q.2: short answers

Q.2 (i) Give any three facts supported by the alpha-particle scattering experiment.

Most alpha particles pass straight through: most of the atom is empty space.
Some alpha particles deflect: positive charge is concentrated in a positively charged nucleus that repels them.
Very few turn through large angles or rebound: the nucleus is very small and dense and contains nearly all the atom's mass.

OR Define specific heat capacity. Calculate heat to raise 25 kg of water by 50 °C.

Specific heat capacity is the energy required to raise the temperature of **1 kg** of a substance by **1 K or 1 °C**, without a change of state.
c = Q/(mΔT), or **Q = mcΔT**.
SI unit: **J kg^{-1} K^{-1}**.
Given: m = 25 kg; ΔT = 50 K. Use c = 4200 J kg^{-1} K^{-1} for water (not printed in the crop).
Q = mcΔT = 25 × 4200 × 50 = 5.25 × 10^{6} J = 5.25 MJ.

Q.2 (ii) Define electric field and give two characteristics.

An electric field is the region in which a charge experiences electric force; **E = F/q**.
1. Direction: field lines point in the force direction on a positive test charge, away from positive charges and towards negative charges.
2. Strength: closer field lines represent a stronger field; wider spacing represents a weaker field.

OR Describe a diode under forward bias and draw a diagram.

Forward bias connects the **p-side/anode towards +** and **n-side/cathode towards −**. The applied voltage reduces the potential barrier and narrows the depletion region. With sufficient forward voltage, majority carriers cross the junction and appreciable current flows. Conventional current in the diode is from p to n. A series resistor limits current.

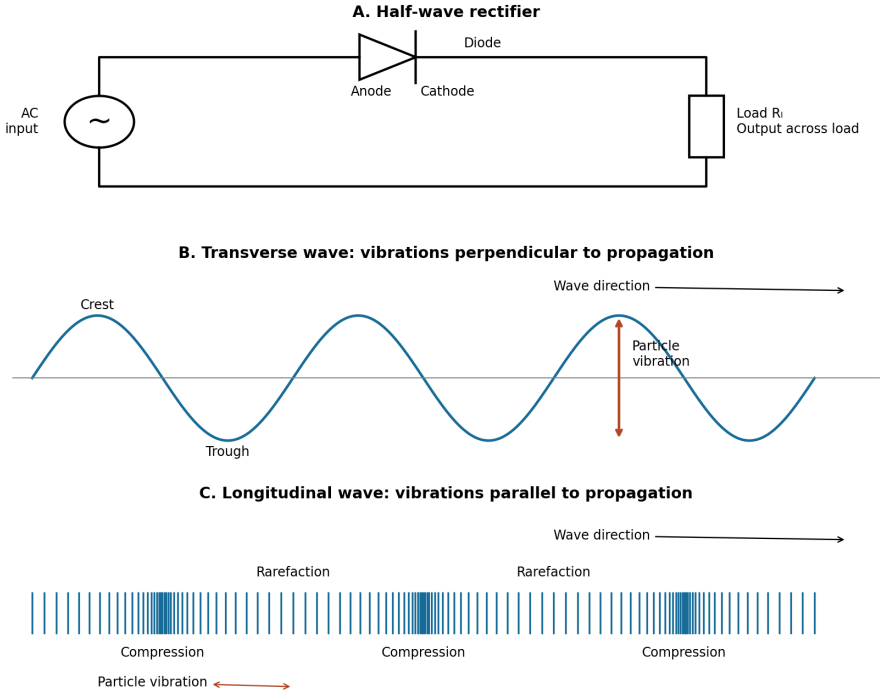
Forward-bias connection: positive supply → current-limiting resistor → p-side (anode) | n-side (cathode) → negative supply. The current path in panel A of the rectifier diagram below is forward biased when its upper source terminal is positive.

Q.2 (iii) Explain with a diagram why transverse vibration is perpendicular to energy transfer.

In a transverse wave, medium particles vibrate **perpendicular** to the direction of energy transfer. For a wave travelling horizontally on a rope, particles move up and down about equilibrium while energy travels horizontally. The particles do not travel with the wave, so there is no net transfer of matter.

OR Find e.m.f. of two 3 V cells in series and parallel.

Series (same sense): E = 3 + 3 = **6 V**.
Parallel (identical cells, like terminals together): E = **3 V**.



Panel A supplies a diode circuit; panel B shows the transverse-wave directions required here.

Q.2 (iv) Find the bill for a 2 kW electric iron used for 3 hours at Rs. 25 per unit.

Given: P = 2 kW; t = 3 h; rate = Rs. 25/kWh
Formula: E = Pt; bill = E × rate
Working: E = 2 × 3 = 6 kWh; bill = 6 × 25
Answer: Rs. 150 (energy charge only).

OR State Snell's law and its mathematical expression.

For a fixed pair of transparent media and wavelength, the ratio of sin i to sin r is constant:
$\sin i / \sin r = n_2/n_1$, or **$n_1 \sin i = n_2 \sin r$**.
Angles are measured from the normal; n_1 and n_2 are the refractive indices of the incident and second media.

Q.2 (v) State Lenz’s law and explain, with an example, its consistency with conservation of energy.

Lenz’s law states that the induced current’s magnetic field opposes the change in magnetic flux producing it. When a north pole approaches a coil, its near face becomes north and repels the magnet. External mechanical work must be done against this force; this supplies the electrical energy produced. If induction assisted its own cause, energy could be created without equivalent input, violating conservation of energy.

OR A 220 V appliance draws 1 A. Find energy used in 5 s.

Given: V = 220 V; I = 1 A; t = 5 s
Formula: E = VIt
Working: E = 220 × 1 × 5
Answer: E = 1100 J = 1.1 kJ.

Q.2 (vi) Describe d.c. and a.c. and give one advantage of each.

D.c. flows in one direction only; steady d.c. has constant magnitude. Advantage: fixed polarity suits electronic circuits and battery charging.
A.c. reverses direction periodically and varies in instantaneous value. Advantage: its voltage can be transformed efficiently for economical transmission and distribution.

OR How does convection in seawater support marine life?

Temperature differences create density differences: warmer, less dense seawater tends to rise and colder, denser water sinks. This circulation **transports dissolved oxygen** into deeper water, **redistributes nutrients** that support aquatic food chains, and **distributes heat** to moderate temperature differences. Ocean circulation also depends on salinity and winds.

Q.2 (vii) Give three differences between pitch and loudness.

Meaning: pitch distinguishes high/shrill from low/deep sound; loudness distinguishes loud from soft sound.
Dependence: pitch mainly depends on frequency; loudness mainly depends on intensity at the ear and hence amplitude.
Change: higher frequency raises pitch; larger amplitude generally increases loudness. Plucking a string harder mainly increases loudness; tightening it raises pitch.

OR Why does heating help an insulator lose charge?

Heating can free trapped charges and increase charge-carrier mobility in an insulator. Its resistance may decrease, allowing accumulated charge to move and leak away through available paths to the surroundings or Earth. If a flame is used, its ions and electrons additionally neutralise the surface charge. Ordinary warming should not be confused with strong ionisation of air.

Q.2 (viii) Write the electromagnetic spectrum from radio waves to gamma rays.

The electromagnetic spectrum is the complete range of electromagnetic waves. In **increasing frequency**:
Radio → Microwaves → Infrared → Visible → Ultraviolet → X-rays → Gamma rays.
Wavelength decreases in this order.

OR State Ohm's law. Find current through 6 Ω with 12 V applied.

At constant temperature and other physical conditions, current through a conductor is directly proportional to the potential difference across it: **$V \propto I$; $V = IR$**.
$I = V/R = 12/6 = 2$ A.

Q.2 (ix) A 50 mg lead-212 sample has half-life 10 h. How much undecayed sample remains after 50 h?

Given: $m_0 = 50$ mg; $T_{½} = 10$ h; $t = 50$ h (use the value supplied by the question)
Formula: $m = m_0(½)^{t/T_{½}}$
Working: Number of half-lives = 50/10 = 5; $m = 50/2^5 = 50/32$
Answer: m = 1.5625 mg of undecayed lead-212; the remainder is decay products.

OR Image height is 10 cm and object height 5 cm. Find magnification and state when it exceeds one.

$|m| = |h_i|/|h_o| = 10/5 = 2$. The image is twice as tall. Orientation is not given, so the sign cannot be assigned. The magnitude exceeds one when image height exceeds object height: **$|h_i| > |h_o|$**. Signed $m > 1$ specifically denotes an upright enlarged image.

Q.2 (x) State image characteristics for an object within a magnifying glass's focal length.

With an object between a convex lens and its principal focus, a magnifying glass produces an image that is **virtual, upright and enlarged**. The image lies on the same side of the lens as the object and cannot be obtained directly on a screen.

OR Find speed for a wave with frequency 50 Hz and wavelength 2 mm in SI units.

Given: $f = 50$ Hz; $\lambda = 2$ mm $= 2 \times 10^{-3}$ m
Formula: $v = f\lambda$
Working: $v = 50 \times 2 \times 10^{-3}$
Answer: v = 0.10 m/s.

Q.2 (xi) State three differences between boiling and evaporation.

Temperature: boiling occurs at a definite boiling point for a given pressure; evaporation can occur at any temperature at which the substance is liquid.
Location: boiling occurs throughout the liquid; evaporation occurs at its surface.
Bubbles: boiling forms vapour bubbles within the liquid; evaporation occurs without such bubbles.

OR Why does the sky change colour at sunset? Name and explain the phenomenon.

The phenomenon is **scattering of light**. At sunset sunlight passes through a longer atmospheric path. Shorter blue/violet wavelengths are scattered out of the direct beam more strongly, leaving a larger proportion of red and orange light from the Sun and nearby horizon. Rayleigh scattering varies approximately as **$1/\lambda^4$**.

Section C | Long answers

Q.3 Explain the greenhouse effect, whether it is good or bad, and its consequences.

Mechanism: Much incoming shortwave sunlight passes through the atmosphere and warms the surface. Earth emits longwave infrared radiation. Greenhouse gases (water vapour, carbon dioxide and methane) absorb some outgoing infrared and re-emit it in all directions, including towards the surface. This reduces the rate of energy loss to space and warms the surface/lower atmosphere.

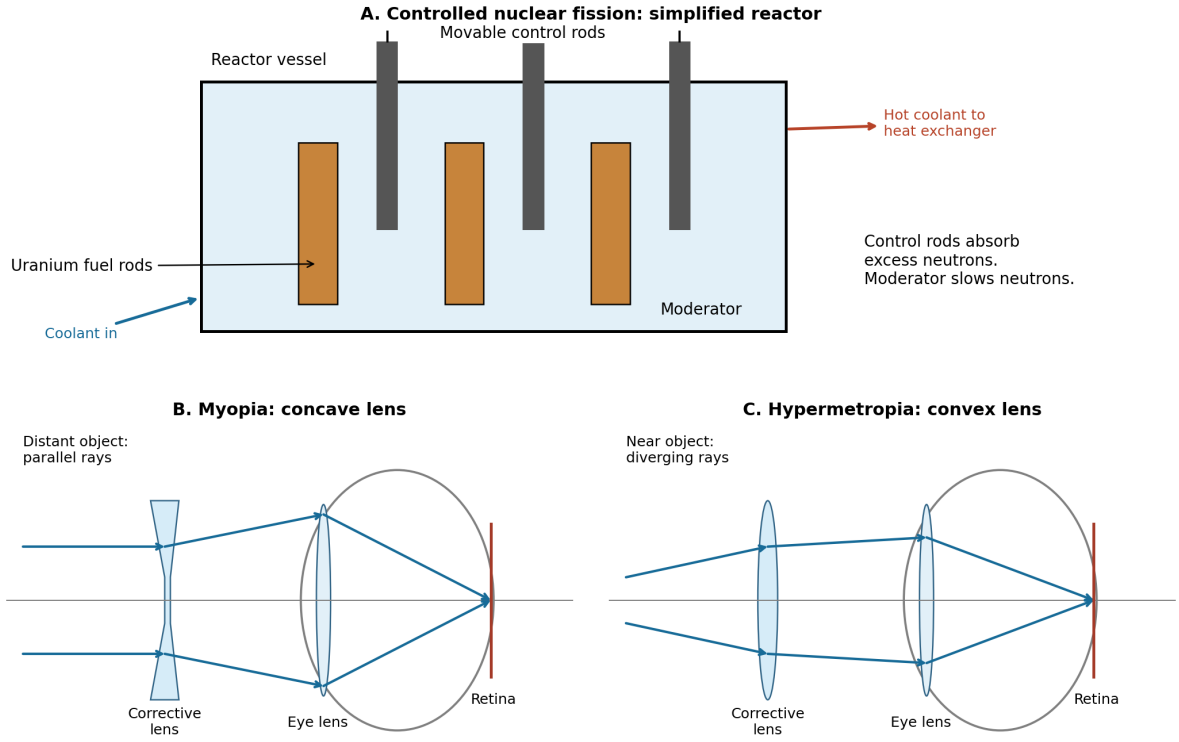
Good or bad? The natural greenhouse effect is beneficial: it keeps Earth warm enough for liquid water and life. Increasing greenhouse-gas concentrations enhance this effect and cause additional warming and harmful climate changes.

Consequences of enhancement:
1. Rising temperatures melt land ice and expand seawater, increasing sea level and coastal flooding risks.
2. Changes in rainfall and more intense heat extremes can increase drought/heavy-rainfall risks, affect agriculture and disrupt ecosystems. The greenhouse effect is not the same as ozone depletion.

Q.3 OR Define nuclear fission and explain controlled fission in a nuclear reactor, with diagram.

Nuclear fission: a heavy nucleus splits after absorbing a neutron, releasing lighter nuclei, energy and more neutrons. Example:
$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + \text{energy.}$$

The released neutrons sustain a chain reaction. In a thermal reactor, a moderator (water or graphite) slows neutrons. Control rods containing boron or cadmium absorb excess neutrons; their position maintains on average one further fission per fission. Inserting rods reduces the rate; withdrawing them increases it. A coolant removes heat, which generates steam to drive a turbine and generator. The steady controlled reaction releases usable energy without a rapidly growing chain reaction.



Panel A shows reactor control; panels B and C show corrected vision for Q.4 OR.

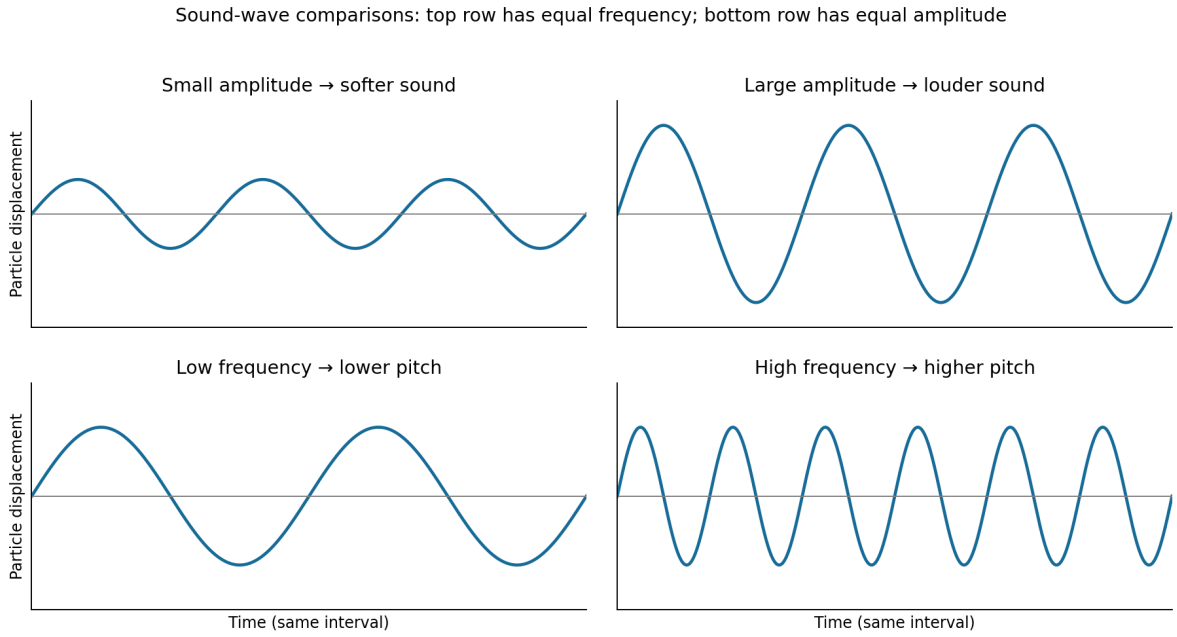
Q.4 (a) Explain amplitude and frequency changes with diagrams. (b) How does frequency determine pitch?

Loudness distinguishes loud from soft sound and depends mainly on intensity reaching the ear. For the same medium and frequency, $I \propto A^2$. Increasing amplitude makes sound generally louder; reducing it makes sound softer. Distance and the ear's sensitivity also influence perceived loudness.

Pitch distinguishes high/shrill from low/deep sound and depends mainly on frequency. More vibrations per second produce a higher pitch; fewer produce a lower pitch. Thus a 1000 Hz tone has a higher pitch than a 200 Hz tone.

Examples: Hit a drum harder or pluck a string more strongly: amplitude and loudness increase, while pitch remains approximately unchanged. Tighten the same string: vibration frequency and pitch increase.

If amplitude doubles, intensity becomes four times as large: $I_2/I_1 = (2A/A)^2 = 4$. Perceived loudness does not necessarily become four times greater. In the diagrams, the top pair keeps frequency constant; the bottom pair keeps amplitude constant. These are displacement-time plots, not drawings of transverse sound propagation in air.



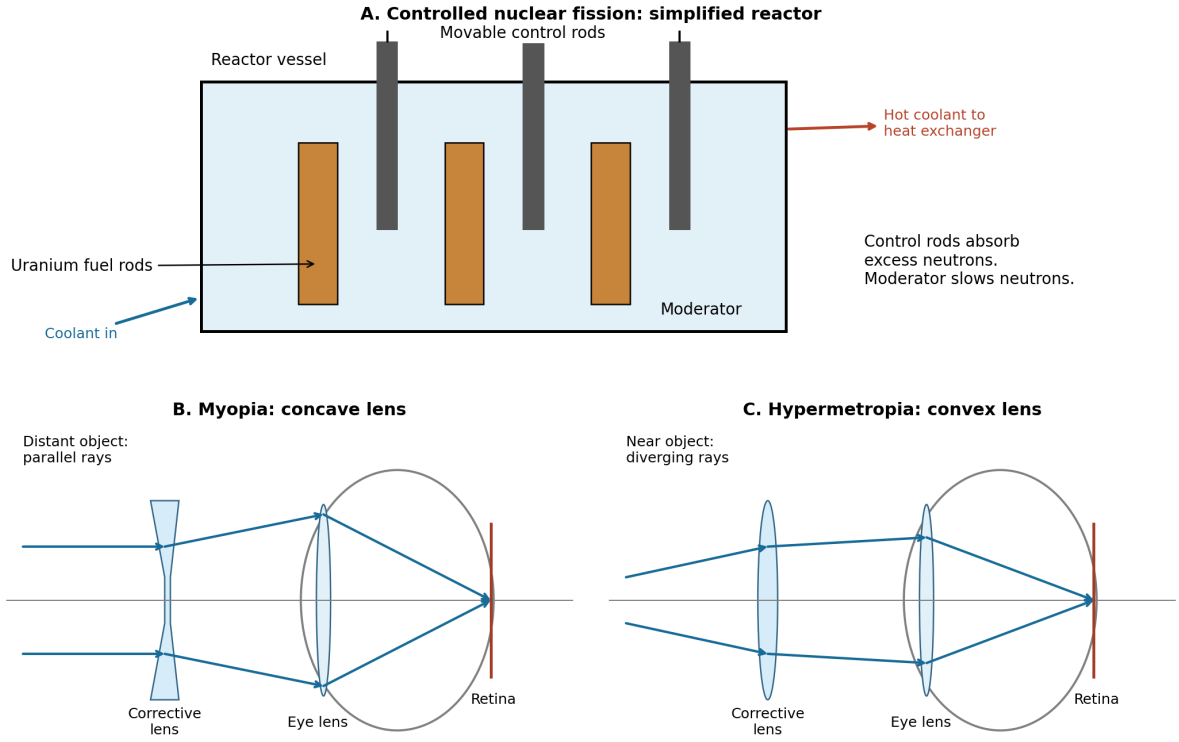
Top pair: equal frequency, different amplitude. Bottom pair: equal amplitude, different frequency.

Q.4 OR Describe lens correction of long-sightedness and short-sightedness, with diagrams.

Long-sightedness (hypermetropia): Near objects are difficult to see clearly. The eyeball may be too short or the eye's converging power insufficient, so near-object rays would focus **behind the retina**. A **convex (converging) lens** reduces their divergence and provides additional convergence, allowing the eye to focus them on the retina.

Short-sightedness (myopia): Distant objects are blurred but nearby objects are clear. The eyeball may be too long or its converging power excessive, so distant parallel rays focus **in front of the retina**. A **concave (diverging) lens** spreads the incoming rays slightly so the eye focuses them farther back, on the retina.

Remember: long-sighted → convex; short-sighted → concave. The labelled diagrams show the corrected rays meeting on the retina.



Panel B: myopia corrected with concave lens. Panel C: hypermetropia corrected with convex lens. Rays meet at the retina.

Q.5 Explain real and apparent thermal expansion of liquids.

Thermal expansion of a liquid is the increase in its volume with temperature. A liquid has no fixed shape, so its expansion is treated as volume expansion. Both the liquid and its container normally expand when heated.

Real expansion: the actual volume increase of the liquid, independent of container expansion. $\mathbf{\Delta V_{real} = \beta_l V_0 \Delta T.}$

Apparent expansion: the observed volume increase relative to the expanding container. The increased container capacity accommodates part of the liquid's true expansion.
$\mathbf{\Delta V_{apparent} = \Delta V_{real} - \Delta V_{container}.}$
Therefore $\mathbf{\Delta V_{real} = \Delta V_{apparent} + \Delta V_{container}}$, and, for the same V_0 and ΔT, $\mathbf{\beta_{apparent} = \beta_l - \beta_c}$.

Flask explanation: Heat a liquid in a flask with a narrow tube. The flask may initially warm and expand first, making the liquid level fall. As the liquid warms it expands and the level rises. The final rise above the original level represents apparent expansion; real expansion also includes the increase in flask capacity. For an ordinarily expanding liquid and container, apparent expansion is smaller than real expansion.

Q.5 OR (a) Derive equivalent resistance for two parallel resistors. (b) Calculate it for two identical 10 Ω resistors.

Connect R_1 and R_2 across the same two nodes. The potential difference is the same across each: $\mathbf{V_1 = V_2 = V}$. Total current divides: $\mathbf{I = I_1 + I_2}$.

By Ohm's law, $I = V/R_p$, $I_1 = V/R_1$ and $I_2 = V/R_2$. Substitution gives:
$\mathbf{V/R_p = V/R_1 + V/R_2.}$
Dividing by V gives $\mathbf{1/R_p = 1/R_1 + 1/R_2}$.
Taking a common denominator:

$1/R_p = (R_1 + R_2)/(R_1R_2)$.
Therefore $\mathbf{R_p = R_1R_2/(R_1 + R_2)}$.

The equivalent resistance is smaller than either positive individual resistance.

Given: $R_1 = R_2 = 10\ \Omega$
Formula: $R_p = R_1R_2/(R_1 + R_2)$
Working: $R_p = (10 \times 10)/(10 + 10) = 100/20$
Answer: $R_p = 5\ \Omega$.

Q.6 (a) Explain step-up and step-down transformers. (b) A transformer converts 240 V to 12 V; primary turns = 2000. Find secondary turns.

Principle: mutual electromagnetic induction. A changing current in the primary creates changing magnetic flux that induces e.m.f. in the secondary.

Construction: Two insulated copper-wire coils are wound on a closed laminated soft-iron core. The primary connects to the a.c. input and the secondary to the load. Coils are electrically insulated from each other and the core. The core guides flux; its laminations reduce eddy-current losses.

Working: Alternating current in the primary creates alternating magnetic flux in the core. Flux links the secondary, inducing an alternating e.m.f. When a load closes the secondary circuit, current flows. A steady d.c. supply cannot sustain transformer action.

Turns relation: $V_s/V_p = N_s/N_p$ (ideal transformer). $N_s > N_p$ gives a step-up transformer; $N_s < N_p$ gives a step-down transformer. At equal ideal power, $\mathbf{V_pI_p = V_sI_s}$, so increasing voltage reduces current.

Energy transfer: electrical input → changing magnetic field → electrical output. The transformer changes voltage/current levels; it does not create energy. Real devices lose some energy as heat in the windings and core.

Given: $V_p = 240$ V; $V_s = 12$ V; $N_p = 2000$
Formula: $N_s = N_p(V_s/V_p)$
Working: $N_s = 2000 \times 12/240$
Answer: $N_s = 100$ turns.

Q.6 OR Explain a burglar alarm and draw its circuit diagram.

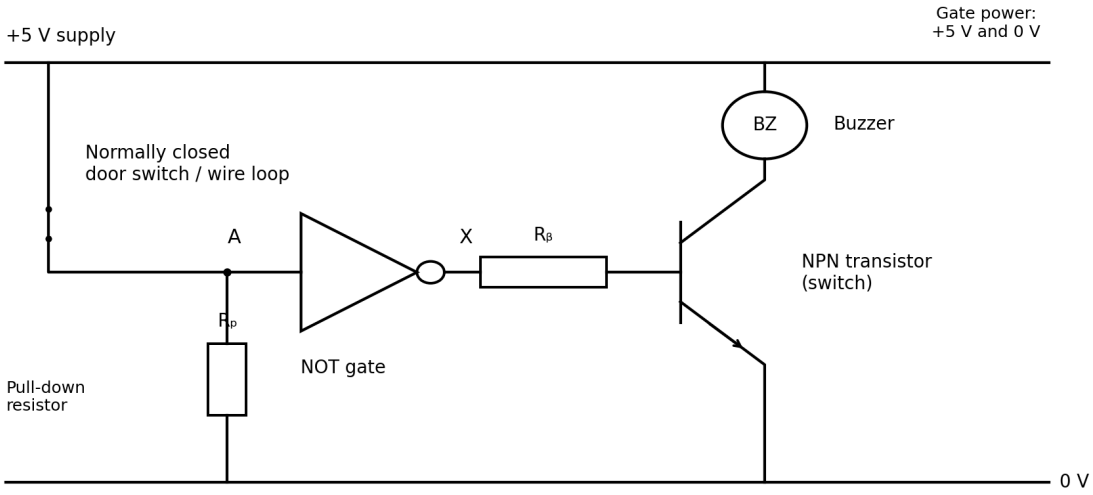
A simple burglar alarm uses a **NOT gate**, a normally closed door-switch/wire loop, a pull-down resistor and a transistor-driven buzzer. Its logic is **X = NOT A**.

Door closed / loop intact: the switch connects input A to the positive supply, so A = 1. The NOT output X = 0. The transistor is off and the buzzer is silent.

Door opened / loop broken: the positive connection opens. The pull-down resistor makes A = 0; the NOT gate gives X = 1. Current through the base resistor turns the NPN transistor on. Current now flows through the buzzer and the alarm sounds.

The pull-down resistor prevents a floating input; the base resistor limits transistor base current. The transistor lets a low-power logic output switch the larger buzzer current. The diagram uses one suitable door-loop design; other sensor arrangements are possible. Supply connections to the logic gate are indicated separately.

Burglar alarm: normally closed door loop + NOT gate + transistor driver



Door closed: A = 1, X = 0, buzzer off. Door opened: A = 0, X = 1, transistor and buzzer on.

Door / loop	Input A	NOT output X	Alarm
Closed / intact	1	0	OFF
Open / broken	0	1	ON

Source map and final study checks

The solution follows the supplied image order. Every visible OR alternative has its own answer; repeated topics are answered again within each paper so the sets remain usable separately.

Paper	Source images (time suffixes identify uploaded crops)
1. 2nd Annual	1(1).jpeg; 2.jpeg
2. 1st Annual - Set 1	121749, 121914 (MCQs); 121950, 122312, 122453 (B); 122553, 122824 (C)
3. 1st Annual - Set 2	123030, 123225 (MCQs); 123308, 123635 (B); 125244, 125434 (C)
4. 1st Annual - Set 3	125713, 130015 (MCQs); 130109, 130221, 130341 (B); 130452, 130809 (C)

Still required for complete source coverage: Paper 1 Section A, and Paper 3 Section B (ix)–(xi), including their OR options.

• Convert mm to m and W to kW when required. One kWh is one electricity unit and equals 3.6×10^6 J.
• Critical and refraction angles are measured from the normal. Total internal reflection requires i greater than C, not equal to C.
• Keep frequency unchanged when waves cross a stationary boundary; speed and wavelength can change.
• State a real-image assumption if an unsigned lens distance leaves the image type unclear.
• Use the half-life specified in the question for its numerical calculation. The remaining parent isotope is not the whole mass of all decay products.
• Use correct current and field directions; a commutator reverses coil current while brushes maintain contact.